

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	21
Flg Calc [Int P]: FLANGE	22
Flg Calc [Int P]: FLANGE	28
Internal Pressure Calculations:	34
Element: CH. COVER	35
Element: CH. SHELL	37
Element: SHELL-1	39
Element: SHELL-2	41
Element: K-SHELL	43
Element: SH. COVER	45
Shop Pressure Test Results:	47
ASME TS Calc: Pressure Test	49
External Pressure Calculations:	57
Element and Detail Weights:	58
Nozzle Flange MAWP:	61
Wind Load Calculation:	62
Earthquake Load Calculation:	64
Center of Gravity Calculation:	66
Saddle Calcs: Bundle Pull New:	67
Saddle Calcs: Bundle Pull Cor:	68
Saddle Calcs: Operating Case:	69
Saddle Calcs: Test Case:	82
Saddle Calcs: Ext. Pressure:	95
Conical Section: SHELL-2	107
Nozzle Calcs.: TS IN NOZ. A1	114
Nozzle Calcs.: TS OUT NOZ. B1	125
Nozzle Calcs.: SS IN NOZ A2	136
Nozzle Calcs.: SS OUT NOZ B2	144
Nozzle Calcs.: SS DRAIN NOZ D1	154
Nozzle Calcs.: TEMP INST. T1	157
Nozzle Calcs.: LEVEL INST L1	160
Nozzle Calcs.: LEVEL INST L2	163
Nozzle Calcs.: LEVEL INST L3	166
Nozzle Calcs.: LEVEL INST L4	169
ASME TS Calc:	172



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRL31EQ0010FFGAS CHILLER.pvdb

Date of Analysis : Sep 8, 2026 11:24am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 410.000 x 850.000 mm. ID
 Vessel Design Length, Tangent to Tangent 4260.98 mm.
 Specified Datum Line Distance 50.00 mm.
 Shell Side Design Temperature 85 °C
 Channel Side Design Temperature 85 °C
 Shell Side Design Pressure 25.000 bars
 Channel Side Design Pressure 25.000 bars
 Shop Shell Side Test Pressure 37.670 bars
 Shop Channel Side Test Pressure 43.807 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N+IT
Head	SA-516 70	...	...	K02700	+N+IT
Cone	SA-516 70	...	...	K02700	+N+IT
Flange	SA-350 LF2	1	...	K03011	+N+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Re-Pad	SA-516 70	...	...	K02700	+IT
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Tubes	SA-334 6	...	...	K03006	+IT
Tubesheet	SA-350 LF2	2	...	K03011	+N+IT
Flg Bolting	SA-320 L7	...	<= 65	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH. COVER	25.027	0.00	70.000	1.5000	No	No
CH. SHELL	25.027	0.00	68.000	1.5000	N/A	No
CH. FLG	25.040	0.00	34.000	1.5000	N/A	No
SH. FLG	25.040	0.00	34.000	1.5000	N/A	No
SHELL-1	25.027	0.00	68.000	1.5000	N/A	No
SHELL-2	25.057	1.03	29.000	1.5000	N/A	No
K-SHELL	25.057	1.03	33.000	1.5000	N/A	No
SH. COVER	25.057	1.03	34.000	1.5000	No	No

Liquid Level: 850.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.681

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	0.0	50.0	14.0	12.0	5.2	...	1.00	1.00



Cylinder	300.0	300.0	12.0	12.0	5.3	...	1.00	1.00
Body Flg	373.2	70.0	60.0	55.0	47.7	47.7	1.00	1.00
Body Flg	498.2	70.0	60.0	55.0	47.7	47.7	1.00	1.00
Cylinder	598.2	100.0	12.0	12.0	5.3	...	1.00	1.00
Conical	1360.2	762.0	12.0	12.0	10.5	4.2	1.00	1.00
Cylinder	4160.2	2800.0	14.0	12.0	9.3	6.1	1.00	1.00
Ellipse	4211.0	50.8	14.0	12.0	9.2	3.8	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
TS IN NOZ. A1	WRC-297	0.362	Shell	Passed
TS OUT NOZ. B1	WRC-297	0.362	Shell	Passed
SS IN NOZ A2	WRC-297	0.739	Shell	Passed
SS OUT NOZ B2	WRC-297	0.981	Shell	Passed

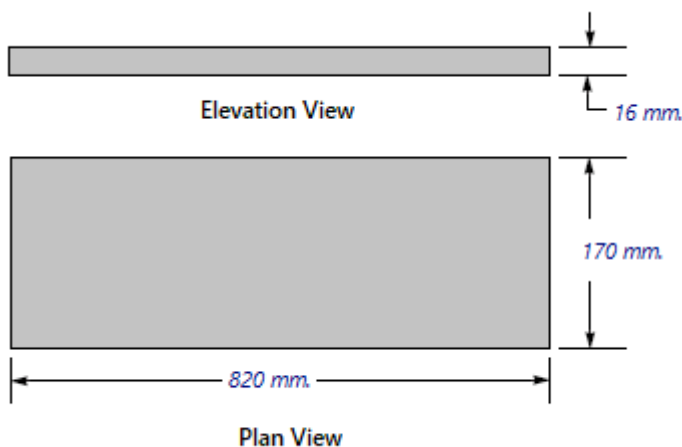
Foundation/Support Summary

Saddle Parameters:

Saddle Width	140.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	700.000 mm.
Wear Pad Width	180.000 mm.
Wear Pad Thickness	12.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	150.000 mm.

Baseplate Length	820.000 mm.
Baseplate Thickness	16.000 mm.
Baseplate Width	170.000 mm.
Number of Ribs (including outside ribs)	3
Rib Thickness	12.000 mm.
Web Thickness	12.000 mm.
Height of Center Web	306.000 mm.

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	3000.38 Kg
Maximum Transverse Saddle Shear Load	285.76 Kg
Maximum Longitudinal Saddle Shear Load	337.85 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	2800.88 Kg
Maximum Transverse Saddle Shear Load	94.30 Kg
Maximum Longitudinal Saddle Shear Load	68.32 Kg



Weights:

Fabricated - Bare w/o removable internals	2387.2 kg.
Shop Test - Fabricated + water (full)	4361.3 kg.
Shipping - Fab. + rem. intls.+ shipping app.	2423.4 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	2423.4 kg.
Empty - Fab. + intls. + details + wghts.	2423.4 kg.
Operating - Empty + operating liquid (no ca)	3773.3 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
TS IN NOZ. A1	68	...	...	OK	5.28	OK	No Calc[*]
TS OUT NOZ. B1	68	...	...	OK	4.92	OK	No Calc[*]
SS IN NOZ A2	33	...	...	OK	6.52	OK	No Calc[*]
SS OUT NOZ B2	33	OK	...	OK	9.31	OK	Passed
SS DRAIN NOZ D1	33	...	...	OK	4.92	OK	No Calc[*]
TEMP INST. T1	33	...	...	OK	4.92	OK	No Calc[*]
LEVEL INST L1	34	...	...	OK	4.72	OK	No Calc[*]
LEVEL INST L2	34	...	...	OK	4.72	OK	No Calc[*]
LEVEL INST L3	34	...	...	OK	4.72	OK	No Calc[*]
LEVEL INST L4	34	...	...	OK	4.72	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 33 Nozzle : SS IN NOZ A2 [Shellside]
 Minimum MAWP Nozzles : 68 Nozzle : TS IN NOZ. A1 [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
20	TS IN NOZ. A1	200.000	90.000	139.299
20	TS OUT NOZ. B1	200.000	-90.000	91.700
70	SS IN NOZ A2	3910.175	90.000	158.400
70	SS OUT NOZ B2	2810.175	90.000	412.400
70	SS DRAIN NOZ D1	2810.175	-90.000	91.700
70	TEMP INST. T1	4010.175	-90.000	91.700
80	LEVEL INST L1	0.000	-225.000	73.970
80	LEVEL INST L2	0.000	-135.000	73.970
80	LEVEL INST L3	0.000	45.000	73.970
80	LEVEL INST L4	0.000	-45.000	73.970

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
LEVEL INST L1	1.500 in	160	LWN	1.900	7.137	...	...	222.81	300
LEVEL INST L2	1.500 in	160	LWN	1.900	7.137	...	...	222.81	300
LEVEL INST L3	1.500 in	160	LWN	1.900	7.137	...	...	222.81	300
LEVEL INST L4	1.500 in	160	LWN	1.900	7.137	...	...	222.81	300
TS OUT NOZ. B1	2.000 in	160	LWN	2.375	8.738	160.32	12.00	214.23	600
SS DRAIN NOZ D1	2.000 in	160	LWN	2.375	8.738	...	...	213.07	300
TEMP INST. T1	2.000 in	160	LWN	2.375	8.738	...	...	213.07	300
TS IN NOZ. A1	3.000 in	160	WNF	3.500	11.125	188.90	12.00	216.88	600
SS IN NOZ A2	3.000 in Actual		WNF	3.876	11.130	...	...	215.99	600
SS OUT NOZ B2	8.000 in Actual		WNF	9.000	12.700	...	...	255.16	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
LEVEL INST	SA-350 LF2	12.000	10.000	...	...	...
LEVEL INST	SA-350 LF2	12.000	10.000	...	...	...
LEVEL INST	SA-350 LF2	12.000	10.000	...	...	...
LEVEL INST	SA-350 LF2	12.000	10.000	...	...	...
TS OUT NOZ.	SA-333 6	12.000	10.000	10.000	12.000	...
SS DRAIN NO	SA-350 LF2	12.000	10.000	...	...	...
TEMP INST.	SA-350 LF2	12.000	10.000	...	...	...
TS IN NOZ.	SA-333 6	12.000	10.000	10.000	12.000	...
SS IN NOZ A	SA-350 LF2	12.000	10.000	...	...	...
SS OUT NOZ	SA-350 LF2	12.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
LEVEL INST L1	...	-225.0	200.00	0.00	SH. COVER
LEVEL INST L2	...	-135.0	200.00	0.00	SH. COVER
LEVEL INST L3	...	45.0	200.00	0.00	SH. COVER
LEVEL INST L4	...	-45.0	200.00	0.00	SH. COVER
TS OUT NOZ. B1	150.000	-90.0	200.00	0.00	CH. SHELL
SS DRAIN NOZ D1	2760.175	-90.0	200.00	0.00	K-SHELL
TEMP INST. T1	3960.175	-90.0	200.00	0.00	K-SHELL
TS IN NOZ. A1	150.000	90.0	200.00	0.00	CH. SHELL
SS IN NOZ A2	3860.175	90.0	200.00	0.00	K-SHELL
SS OUT NOZ B2	2760.175	90.0	200.00	0.00	K-SHELL

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm. THK X 410.0mm. ID X 50.0mm.	SA-516 70
1	CYLINDER: 12.0mm. THK X 410.0mm. ID X 300.0mm.	SA-516 70
2	WELD NECK FLANGE: 60.0mm. THK X 538.0mm. OD	SA-350 LF2
1	CYLINDER: 12.0mm. THK X 410.0mm. ID X 100.0mm.	SA-516 70
1	ECCENTRIC CONE: 12.0mm. THK X 850.0mm. X 410.0mm. ID X 762.0 mm.	SA-516 70
1	CYLINDER: 14.0mm. THK X 850.0mm. ID X 2800.0mm.	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 14.0mm. THK X 850.0mm. ID X 50.8mm.	SA-516 70
1	INSULATION: 152mm X 70mm THK	COLD INSULATION
1	INSULATION: 300mm X 70mm THK	COLD INSULATION
2	INSULATION: 69mm X 70mm THK	COLD INSULATION
5	BAFFLES: 390.0mm DIA X 6.0mm THK	SA-516 70
1	TUBESHEET: 538.0mm X 55.0mm THK	SA-350 LF2
147	TUBES: 3600.0mm X 19.0mm DIA X 2.1mm THK	SA-334 6
1	INSULATION: 100mm X 70mm THK	COLD INSULATION
1	INSULATION: 762mm X 70mm THK	COLD INSULATION
2	SADDLE: 140.0mm X 120 DEG	SA-516 70
1	INSULATION: 2800mm X 70mm THK	COLD INSULATION
1	INSULATION: 263mm X 70mm THK	COLD INSULATION
2	CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)	SA-350 LF2
4	CLASS 300 GR 1.1, 1.5" BLIND FLANGE(S)	SA-350 LF2
1	GASKET: 439.0mm. OD X 413.0mm. ID	...
16	BODY FLANGE BOLTS: 20.0mm. DIA	SA-320 L7
32	NUTS FOR BODY FLANGE BOLTS: 20.0mm. DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
SH. FLG	[11]	!	-46	-46		0.426	10.000	1.00	No
SHELL-1	[8]	!	-40	-86		0.418	12.000	1.00	No
SHELL-2	[8]	!	-40	-40		0.998	12.000	1.00	No
K-SHELL	[8]	!	-40	-48		0.864	12.000	1.00	No
SH. COVER	[10]	!	-40	-48		0.851	12.000	1.00	No
SH. COVER	[7]	!	-40	-55		0.726	14.000	1.00	No
SS IN NOZ A2	[1]	!	-46	-46		0.744	12.000	1.00	No
Nozzle Flg	[5]	!	-46	-104					
SS OUT NOZ B2	[1]	!	-46	-46		0.744	12.000	1.00	No
Nozzle Flg	[5]	!	-46	-79					
SS DRAIN NOZ D	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
TEMP INST. T1	[1]	!	-46	-104		0.068	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
LEVEL INST L1	[1]	!	-46	-104		0.071	6.245	1.00	No
Nozzle Flg	[5]	!	-46	-46					
LEVEL INST L2	[1]	!	-46	-104		0.072	6.245	1.00	No
Nozzle Flg	[5]	!	-46	-46					
LEVEL INST L3	[1]	!	-46	-104		0.071	6.245	1.00	No
Nozzle Flg	[5]	!	-46	-46					
LEVEL INST L4	[1]	!	-46	-104		0.072	6.245	1.00	No
Nozzle Flg	[5]	!	-46	-46					
Tubesheet: SS	[13]	!	-18	-18	-29	0.741	13.750	1.00	No
Warmest MDMT:			-18	-18					
CH. FLG	[11]	!	-46	-46		0.495	10.000	1.00	No
CH. COVER	[10]	!	-40	-76		0.477	12.000	1.00	No
CH. COVER	[7]	!	-40	-88		0.409	14.000	1.00	No
CH. SHELL	[8]	!	-40	-74		0.487	12.000	1.00	No
TS IN NOZ. A1	[1]	!	-46	-104		0.091	9.735	1.00	No
Nozzle Flg	[5]	!	-46	-104					
TS OUT NOZ. B1	[1]	!	-46	-104		0.080	7.645	1.00	No
Nozzle Flg	[5]	!	-46	-46					
Tubesheet: CS	[14]	!	-18	-18	-29	0.741	13.750	1.00	No
Bolting	[21]		-101						
Warmest MDMT:			-18	-18					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	-40.0	Fail
Channel/Tube	-29.0	-40.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)



[20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
[21] - Body Flange Bolting Material
[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfcmm2mm):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	25	bars
Channel Side Design Pressure	25	bars
Shell Side Design Temperature	85.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Amine	
Service Type, Channel Side	Sour	
MDMT (CET), Shell Side	-40.0	°C
MDMT (CET), Tube Side	-40.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS



Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	300 mm.
Percent Wind for Hydrottest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No



Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/8
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	CH. COVER
Distance "FROM" to "TO"	50 mm.
Inside Diameter	410 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	410 mm.
Liquid Density	0.0006749 kg./cm ³

Element From Node	10
Detail Type	Insulation
Detail ID	Ins: 10
Dist. from "FROM" Node / Offset dist	-102.5 mm.
Height/Length of Insulation	152.5 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm ³

Element#	2/8
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	CH. SHELL
Distance "FROM" to "TO"	300 mm.
Inside Diameter	410 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	12 mm.



External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	410 mm.
Liquid Density	0.0006749 kg./cm³

Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	300 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm³

Element From Node	20
Detail Type	Nozzle
Detail ID	TS IN NOZ. A1
Dist. from "FROM" Node / Offset dist	150 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	14.678 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT

Element From Node	20
Detail Type	Nozzle
Detail ID	TS OUT NOZ. B1
Dist. from "FROM" Node / Offset dist	150 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	-90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	8.9148 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT

Element#	3/8
Element From Node	30
Element To Node	40
Element Type	Flange
Description	CH. FLG
Distance "FROM" to "TO"	70 mm.
Flange Inside Diameter	410 mm.
Element Thickness	55 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	60 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2



Material Name	SA-350 LF2 +N+IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	229.6 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	1 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	410 mm.
Liquid Density	0.0009984 kg./cm ³
Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	70 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm ³

Element#	4/8
Element From Node	40
Element To Node	50
Element Type	Flange
Description	SH. FLG
Distance "FROM" to "TO"	70 mm.
Flange Inside Diameter	410 mm.
Element Thickness	55 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	60 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +N+IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	410 mm.
Liquid Density	0.0009984 kg./cm ³
Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	70 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm ³



Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	SHELL-1
Distance "FROM" to "TO"	100 mm.
Inside Diameter	410 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	262 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	410 mm.
Liquid Density	0.0006799 kg./cm ³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	100 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm ³

Element#	6/8
Element From Node	60
Element To Node	70
Element Type	Conical
Description	SHELL-2
Distance "FROM" to "TO"	762 mm.
Inside Diameter	410 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Cone Diameter at "To" End	850 mm.
Design Length of Cone	762 mm.
Half Apex Angle of Cone	30.0 degrees
Toriconical (Y/N)	N



Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	850 mm.
Liquid Density	0.0006799 kg./cm ³
Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	762 mm.
Thickness of Insulation	70 mm.
Density	0.40000E-04 kg./cm ³

Element#	7/8
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	K-SHELL
Distance "FROM" to "TO"	2800 mm.
Inside Diameter	850 mm.
Element Thickness	12 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	25 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Saddle
Detail ID	SLIDING SADDLE
Dist. from "FROM" Node / Offset dist	150 mm.
Width of Saddle	140 mm.
Height of Saddle at Bottom	700 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	180 mm.
Thickness of Wear Plate	12 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.1
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	70
Detail Type	Saddle
Detail ID	FIXED SADDLE
Dist. from "FROM" Node / Offset dist	2240 mm.
Width of Saddle	140 mm.
Height of Saddle at Bottom	700 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	180 mm.
Thickness of Wear Plate	12 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.60000002



Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid 70	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	850	mm.
Liquid Density	0.0006799	kg./cm ³
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	2800	mm.
Thickness of Insulation	70	mm.
Density	0.40000E-04	kg./cm ³
Element From Node	70	
Detail Type	Nozzle	
Detail ID	SS IN NOZ A2	
Dist. from "FROM" Node / Offset dist	2500	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	16.945	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	SS OUT NOZ B2	
Dist. from "FROM" Node / Offset dist	1400	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	63.239	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	SS DRAIN NOZ D1	
Dist. from "FROM" Node / Offset dist	1400	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	-90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	TEMP INST. T1	
Dist. from "FROM" Node / Offset dist	2600	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	-90.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf



Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element# 8/8
 Element From Node 80
 Element To Node 90
 Element Type Elliptical
 Description SH. COVER
 Distance "FROM" to "TO" 50.8 mm.
 Inside Diameter 850 mm.
 Element Thickness 12 mm.
 Internal Corrosion Allowance 1.5 mm.
 Nominal Thickness 14 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 25 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 1.0342 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +N+IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 80
 Detail Type Liquid
 Detail ID Liquid: 80
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 850 mm.
 Liquid Density 0.0006799 kg./cm³

Element From Node 80
 Detail Type Insulation
 Detail ID Ins: 80
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 263.3 mm.
 Thickness of Insulation 70 mm.
 Density 0.40000E-04 kg./cm³

Element From Node 80
 Detail Type Nozzle
 Detail ID LEVEL INST L1
 Dist. from "FROM" Node / Offset dist 282.84 mm.
 Nozzle Diameter 1.5 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle -225.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 0 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 80
 Detail Type Nozzle
 Detail ID LEVEL INST L2
 Dist. from "FROM" Node / Offset dist 282.84 mm.
 Nozzle Diameter 1.5 in.
 Nozzle Schedule 160
 Nozzle Class 300
 Layout Angle -135.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 7.8339 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 80
 Detail Type Nozzle



Detail ID	LEVEL INST L3
Dist. from "FROM" Node / Offset dist	282.84 mm.
Nozzle Diameter	1.5 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	45.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	80
Detail Type	Nozzle
Detail ID	LEVEL INST L4
Dist. from "FROM" Node / Offset dist	282.84 mm.
Nozzle Diameter	1.5 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	-45.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
CH. COVER		50	...	50	...
CH. SHELL		350	...	300	...
CH. FLG		423.175	...	70	...
SH. FLG		548.175	...	70	...
SHELL-1		648.175	...	100	...
SHELL-2		1410.17	...	762	...
K-SHELL		4210.18	...	2800	...
SH. COVER		4260.98	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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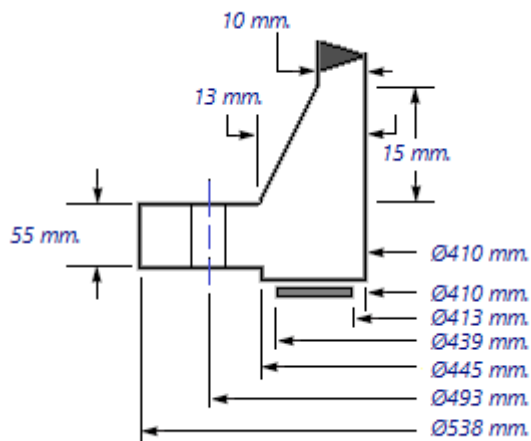


Flange Input Data Values

Description: FLANGE :

CH. FLG

Flange Type	Integral Weld Neck		
Design Pressure	P	25.04	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	1.5000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	410.000	mm.
Flange Outside Diameter	A	538.000	mm.
Flange Thickness	t	55.0000	mm.
Thickness of Hub at Small End	go	10.0000	mm.
Thickness of Hub at Large End	gl	13.0000	mm.
Length of Hub	h	15.0000	mm.
Flange Material	SA-350 LF2 +N+IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-320 L7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	493.000	mm.
Nominal Bolt Diameter	a	20.0000	mm.
Type of Threads	TEMA Metric		
Number of Bolts	16		
Flange Face Outside Diameter	Fod	445.000	mm.
Flange Face Inside Diameter	Fid	410.000	mm.
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	439.000	mm.
Gasket Inside Diameter	Gi	413.000	mm.
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	17.24	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.04 \cdot (410.0/2 + 1.5)) / (138.0 \cdot 1.0 - 0.6 \cdot 25.04) + Ca$$

= 5.2884 mm.

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.5) / (206.5 + 0.6 \cdot 8.5)$$

$$= 55.432 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.0) / (205.0 + 0.6 \cdot 10.0)$$

$$= 65.399 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	413.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	11.500 mm.
Corroded Small Hub,	g0Cor = go-ci	8.500 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	28.500 mm.

Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm.
Basic Gasket Width,	bo = N/2	6.500 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	426.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 426.1509^2 \cdot 25.04$$

$$= 36420.910 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 426.1509 \cdot 2.0 \cdot 25.04$$

$$= 8785.180 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 413.0^2 \cdot 25.0402$$

$$= 34207.727 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 36421 - 34208$$

$$= 2213.185 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(36421 + 8785 + 0, 0)$$

$$= 45206.090 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 17.24 \cdot 6.4246 \cdot \pi \cdot 426.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 15118.270 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((45206 + 0 + 4 \cdot 0 / 426.15) / 172, 15118 / 172)$$

$$= 25.775 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 20.0 + 6 \cdot 55.0 / (2.0 + 0.5)$$

$$= 172.000 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 493.0 \cdot \sin(\pi / 16)$$

$$= 96.180 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{96.18}{2 \cdot 20.0 + 55.0}}, 1)$$

$$= 1.0062$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	25.775	34.728	
Radial Distance between Hub and Bolts:	28.200	28.500	
Radial Distance between Bolts and Edge:	22.500	22.500	
Circ. Spacing between the Bolts:	46.800	96.180	172.000

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 34.728 \cdot 172.0 / (17.24 \cdot \pi(439.0 + 413.0))$$

$$= 12.946 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 34.7282$$

$$= 60907.91 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 45206 - 36421$$

$$= 8785.18 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (493.0 - 426.1509) / 2$$

$$= 33.4246 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((493.0 - 413.0) / 2 + 33.4246)$$

$$= 36.7123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (493.0 - 413.0 - 11.5) / 2$$

$$= 34.2500 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	34208.	34.2500	1.0062	1179.
Face Pressure, Mt	2213.	36.7123	1.0062	82.
Gasket Load, Mg	8785.	33.4246	1.0062	295.
Gasket Seating, Matm	60908.	33.4246	1.0062	2048.

Total Moment for Operation, Mo 1556. Kg-m.
 Total Moment for Gasket seating, Mg 2048. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 59.249 mm.
 Hub Ratio, h/h0 = HL / H0 0.253
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.353

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.896
 Factor V per 4.16.5 0.418
 Factor f per 4.16.5 1.079

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 538.0 / 413.0$$

= 1.303

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.303^2 \cdot \log_{10}(1.303))) / (1.303^2 - 1)) / (1.303 - 1)$$

$$= 7.490$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.303^2(1 + 8.55246 \cdot \log_{10}(1.303)) - 1) / ((1.0472 + 1.9448 \cdot 1.303^2) \cdot (1.303 - 1))$$

$$= 1.796$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.303^2(1 + 8.55246 \cdot \log_{10}(1.303)) - 1) / (1.36136(1.303^2 - 1) \cdot (1.303 - 1))$$

$$= 8.231$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.303^2 + 1) / (1.303^2 - 1)$$

$$= 3.870$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 8.231 \cdot 8.5^2 \cdot 59.249 / 0.418$$

$$= 84280.922 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.896 / 59.249$$

$$= 0.015 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (55.0 \cdot 0.015 + 1) / 1.796 + 55.0^3 / 84280.922$$

$$= 2.994$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0787 \cdot 1556 / 413.0) / (2.9936 \cdot 11.5^2)$$

$$= 100.68 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0787 \cdot 2048 / 413.0) / (2.9936 \cdot 11.5^2)$$

$$= 132.53 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.1084 \cdot 1556 / 413.0) / (2.9936 \cdot 55.0^2)$$

$$= 8.60 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.1084 \cdot 2048 / 413.0) / (2.9936 \cdot 55.0^2)$$

$$= 11.32 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (7.4898 \cdot 1556 / (55.0^2 \cdot 413.0)) - 3.8697 \cdot 9$$

$$= 58.20 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (7.4898 \cdot 2048 / (55.0^2 \cdot 413.0)) - 3.8697 \cdot 11$$

$$= 76.61 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (101 + \max(9, 58)) / 2$$

$$= 79.44 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (133 + \max(11, 77)) / 2$$

$$= 104.57 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 45206 / 34.7282$$

$$= 127.66 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (15118 / 34.7282)$$

$$= 42.69 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	100.68	207.00	132.53 207.00
Radial Flange	8.60	138.00	11.32 138.00
Tangential Flange	58.20	138.00	76.61 138.00
Maximum Average	79.44	138.00	104.57 138.00
Bolting	127.66	172.00	42.69 172.00

Minimum Required Flange Thickness 47.650 mm.

Estimated M.A.W.P. (Operating) 34 bars
 Estimated Finished Weight of Flange at given Thk. 42.4 kg.
 Estimated Unfinished Weight of Forging at given Thk 51.7 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2132.6 / 1.0475 \cdot 9806.637 \cdot 0.418 / (2.24 \cdot 202713 \cdot 8.5^2 \cdot 59.249 \cdot 0.3)$$

$$= 0.746 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1620.0 / 1.0475 \cdot 9806.637 \cdot 0.418 / (2.24 \cdot 199054 \cdot 8.5^2 \cdot 59.249 \cdot 0.3)$$

$$= 0.577 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2048.5 / 1.0062 \cdot 9806.637 \cdot 0.418 / (2.994 \cdot 202713 \cdot 8.5^2 \cdot 59.249 \cdot 0.3)$$

$$= 0.558 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1556.1 / 1.0062 \cdot 9806.637 \cdot 0.418 / (2.994 \cdot 199054 \cdot 8.5^2 \cdot 59.249 \cdot 0.3)$$

$$= 0.432 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:



Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	97.57	207.00	126.11	207.00
Radial Flange	10.58	138.00	13.68	138.00
Tangential Flange	51.52	138.00	66.59	138.00
Maximum Average	74.54	138.00	96.35	138.00
Bolting	127.66	172.00	42.69	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 44.526 mm.

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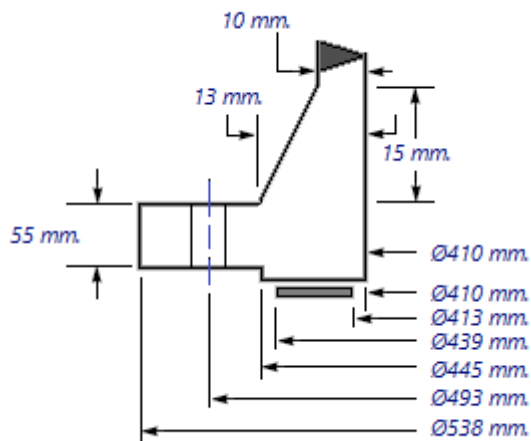


Flange Input Data Values

Description: FLANGE :

SH. FLG

Flange Type	Integral Weld Neck		
Design Pressure	P	25.04	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	1.5000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	410.000	mm.
Flange Outside Diameter	A	538.000	mm.
Flange Thickness	t	55.0000	mm.
Thickness of Hub at Small End	go	10.0000	mm.
Thickness of Hub at Large End	gl	13.0000	mm.
Length of Hub	h	15.0000	mm.
Flange Material	SA-350 LF2 +N+IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-320 L7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	493.000	mm.
Nominal Bolt Diameter	a	20.0000	mm.
Type of Threads	TEMA Metric		
Number of Bolts	16		
Flange Face Outside Diameter	Fod	445.000	mm.
Flange Face Inside Diameter	Fid	410.000	mm.
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	439.000	mm.
Gasket Inside Diameter	Gi	413.000	mm.
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	17.24	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.04 \cdot (410.0/2 + 1.5)) / (138.0 \cdot 1.0 - 0.6 \cdot 25.04) + Ca$$

= 5.2884 mm.

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 8.5) / (206.5 + 0.6 \cdot 8.5) \\ &= 55.432 \text{ bars} \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 10.0) / (205.0 + 0.6 \cdot 10.0) \\ &= 65.399 \text{ bars} \end{aligned}$$

Corroded Flange ID,	Bcor = B+2*ci	413.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	11.500 mm.
Corroded Small Hub,	g0Cor = go-ci	8.500 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	28.500 mm.

Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm.
Basic Gasket Width,	bo = N/2	6.500 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	426.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 426.1509^2 \cdot 25.04 \\ &= 36420.910 \text{ Kg} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\ &= 2 \cdot 6.4246 \cdot \pi \cdot 426.1509 \cdot 2.0 \cdot 25.04 \\ &= 8785.180 \text{ Kg} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 413.0^2 \cdot 25.0402 \\ &= 34207.727 \text{ Kg} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 36421 - 34208 \\ &= 2213.185 \text{ Kg} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(36421 + 8785 + 0, 0) \\ &= 45206.090 \text{ Kg} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 17.24 \cdot 6.4246 \cdot \pi \cdot 426.151 + 0.0 \cdot 0.0 \cdot 0.0 \\ &= 15118.270 \text{ Kg} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\ &= \max((45206 + 0 + 4 \cdot 0 / 426.15) / 172, 15118 / 172) \\ &= 25.775 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t / (m + 0.5) \\ &= 2 \cdot 20.0 + 6 \cdot 55.0 / (2.0 + 0.5) \\ &= 172.000 \text{ mm.} \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned} &= C \cdot \sin(\pi / n) \\ &= 493.0 \cdot \sin(\pi / 16) \\ &= 96.180 \text{ mm.} \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:



$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{96.18}{2 \cdot 20.0 + 55.0}}, 1)$$

$$= 1.0062$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	25.775	34.728	
Radial Distance between Hub and Bolts:	28.200	28.500	
Radial Distance between Bolts and Edge:	22.500	22.500	
Circ. Spacing between the Bolts:	46.800	96.180	172.000

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 34.728 \cdot 172.0 / (17.24 \cdot \pi(439.0 + 413.0))$$

$$= 12.946 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 34.7282$$

$$= 60907.91 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 45206 - 36421$$

$$= 8785.18 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (493.0 - 426.1509) / 2$$

$$= 33.4246 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((493.0 - 413.0) / 2 + 33.4246)$$

$$= 36.7123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (493.0 - 413.0 - 11.5) / 2$$

$$= 34.2500 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	34208.	34.2500	1.0062	1179.
Face Pressure, Mt	2213.	36.7123	1.0062	82.
Gasket Load, Mg	8785.	33.4246	1.0062	295.
Gasket Seating, Matm	60908.	33.4246	1.0062	2048.

Total Moment for Operation, Mo 1556. Kg-m.
 Total Moment for Gasket seating, Mg 2048. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 59.249 mm.
 Hub Ratio, h/h0 = HL / H0 0.253
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.353

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.896
 Factor V per 4.16.5 0.418
 Factor f per 4.16.5 1.079

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 538.0 / 413.0$$

= 1.303

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.303^2 \cdot \log_{10}(1.303))) / (1.303^2 - 1)) / (1.303 - 1)$$

$$= 7.490$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.303^2(1 + 8.55246 \cdot \log_{10}(1.303)) - 1) / ((1.0472 + 1.9448 \cdot 1.303^2) \cdot (1.303 - 1))$$

$$= 1.796$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.303^2(1 + 8.55246 \cdot \log_{10}(1.303)) - 1) / (1.36136(1.303^2 - 1) \cdot (1.303 - 1))$$

$$= 8.231$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.303^2 + 1) / (1.303^2 - 1)$$

$$= 3.870$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 8.231 \cdot 8.5^2 \cdot 59.249 / 0.418$$

$$= 84280.922 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.896 / 59.249$$

$$= 0.015 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (55.0 \cdot 0.015 + 1) / 1.796 + 55.0^3 / 84280.922$$

$$= 2.994$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0787 \cdot 1556 / 413.0) / (2.9936 \cdot 11.5^2)$$

$$= 100.68 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0787 \cdot 2048 / 413.0) / (2.9936 \cdot 11.5^2)$$

$$= 132.53 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.1084 \cdot 1556 / 413.0) / (2.9936 \cdot 55.0^2)$$

$$= 8.60 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.1084 \cdot 2048 / 413.0) / (2.9936 \cdot 55.0^2)$$

$$= 11.32 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (7.4898 \cdot 1556 / (55.0^2 \cdot 413.0)) - 3.8697 \cdot 9$$

$$= 58.20 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (7.4898 \cdot 2048 / (55.0^2 \cdot 413.0)) - 3.8697 \cdot 11$$

$$= 76.61 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (101 + \max(9, 58)) / 2$$

$$= 79.44 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (133 + \max(11, 77)) / 2$$

$$= 104.57 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 45206 / 34.7282$$

$$= 127.66 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (15118 / 34.7282)$$

$$= 42.69 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	100.68	207.00	132.53 207.00
Radial Flange	8.60	138.00	11.32 138.00
Tangential Flange	58.20	138.00	76.61 138.00
Maximum Average	79.44	138.00	104.57 138.00
Bolting	127.66	172.00	42.69 172.00

Minimum Required Flange Thickness 47.650 mm.

Estimated M.A.W.P. (Operating) 34 bars
 Estimated Finished Weight of Flange at given Thk. 42.4 kg.
 Estimated Unfinished Weight of Forging at given Thk 51.7 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2132.6 / 1.0475 \cdot 9806.637 \cdot 0.418 / (2.24 \cdot 202713 \cdot 8.5^2 \cdot 59.249 \cdot 0.3)$$

$$= 0.746 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1620.0 / 1.0475 \cdot 9806.637 \cdot 0.418 / (2.24 \cdot 199054 \cdot 8.5^2 \cdot 59.249 \cdot 0.3)$$

$$= 0.577 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 2048.5 / 1.0062 \cdot 9806.637 \cdot 0.418 / (2.994 \cdot 202713 \cdot 8.5^2 \cdot 59.249 \cdot 0.3)$$

$$= 0.558 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 1556.1 / 1.0062 \cdot 9806.637 \cdot 0.418 / (2.994 \cdot 199054 \cdot 8.5^2 \cdot 59.249 \cdot 0.3)$$

$$= 0.432 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	97.57	207.00	126.11	207.00
Radial Flange	10.58	138.00	13.68	138.00
Tangential Flange	51.52	138.00	66.59	138.00
Maximum Average	74.54	138.00	96.35	138.00
Bolting	127.66	172.00	42.69	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 44.526 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
CH. COVER		25.027	14	1.5	410	138	1
CH. SHELL		25.027	12	1.5	410	138	2
CH. FLG		25.04	60	1.5	410	138	3
SH. FLG		25.04	60	1.5	410	138	4
SHELL-1		25.027	12	1.5	410	138	5
SHELL-2		25.057	12	1.5	850	138	6
K-SHELL		25.057	14	1.5	850	138	7
SH. COVER		25.057	14	1.5	850	138	8

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.W.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
CH. COVER		25	70.455	80.3057	12	5.2161	1
CH. SHELL		25	68.0609	78.035	12	5.28644	2
CH. FLG		25	33.6974	33.7376	55	47.6504	3
SH. FLG		25	33.6974	33.7376	55	47.6504	4
SHELL-1		25	68.0607	78.035	12	5.28648	5
SHELL-2		25	28.9768	33.2546	12	10.546	6
K-SHELL		25	33.4209	38.3134	12	9.32977	7
SH. COVER		25	33.9904	38.8527	12	9.22236	8

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

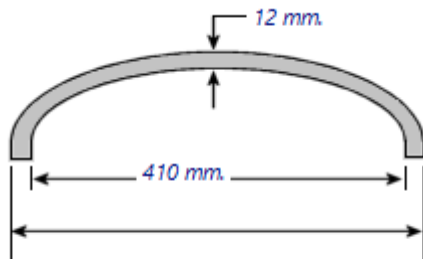
Shell Side MAWP = 29 bars
 Shell Side MAPnc = 33 bars
 Channel Side MAWP = 34 bars
 Channel Side MAPnc = 34 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: CH. COVER

Elliptical Head Design Information:					
Design Pressure	P	25.027 bars	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		50.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	410.000 mm.	Actual Stress	49.0 N./mm ²	
Spec. Min. Thk t _e , or sa		12.0000 mm.	Spec. Nominal Thk.	14.000 mm.	
Surface Area		2767.6 cm ²	SG of Contents	0.676	
Element Volume		15625.8 cm ³	Weight of Contents	10.6 Kg _f	
Empty Weight		30.6 Kg _f	Operating Weight	41.1 Kg _f	
			Comp. All. Ope. Stress	145.7 N./mm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 5.125 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 2.562 mm.

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.027 \cdot 413.0 \cdot 0.99) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.027)$$

$$= 3.7161 + 1.5000 = 5.2161 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.027 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 10.5) / (0.99 \cdot 413.0 + 0.2 \cdot 10.5)$$

$$= 70 - 0 = 70 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 410.0 + 0.2 \cdot 12.0)$$

$$= 80 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.027 \cdot (0.99 \cdot 413.0 + 0.2 \cdot 10.5)) / (2 \cdot 1.0 \cdot 10.5)$$



= 49.002 N./mm²

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (25.027 \cdot 206.5) / (138.0 \cdot 1.0 - 0.6 \cdot 25.027) + 1.5 \\ &= 5.286 \text{ mm.} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.027 bars](#)

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 12.5) / (206.5 + 0.6 \cdot 12.5) \\ &= 81 - 0 = 81 \text{ bars} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (413.0 / (2 \cdot 104.0)))^2 / 6 \\ &= 0.990419 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 13.690 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

[Note: This Element/Detail was specified as being Impact Tested.](#)

MDMT Calculations in the Head Straight Flange:

[Note: This Element/Detail was specified as being Impact Tested.](#)

Tolerance for Formed Heads per UG-81(a):

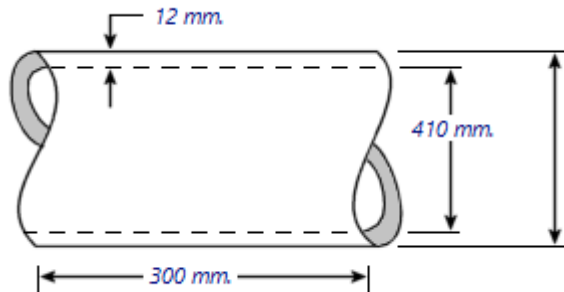
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 5.125 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 2.562 mm.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: CH. SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	25.027 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		300.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	410.000 mm.	Actual Stress		50.7 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		4090.4 cm ²	SG of Contents		0.676
Element Volume		39614.7 cm ³	Weight of Contents		26.8 Kg
Empty Weight		37.0 Kg	Operating Weight		63.8 Kg
			Comp. All. Ope. Stress		145.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.027 \cdot 206.5) / (138.0 \cdot 1.0 - 0.6 \cdot 25.027)$$

$$= 3.7864 + 1.5000 = 5.2864 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.027 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.5) / (206.5 + 0.6 \cdot 10.5)$$

$$= 68 - 0 = 68 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (205.0 + 0.6 \cdot 12.0)$$

$$= 78 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.027 \cdot (206.5 + 0.6 \cdot 10.5)) / (1.0 \cdot 10.5)$$

$$= 50.725 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.844 %

Minimum Design Metal Temperature Results:



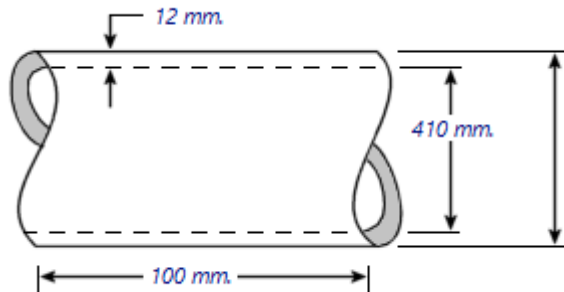
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	25.027 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		100.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	410.000 mm.	Actual Stress		50.7 N./mm ²
Spec. Min. Thk t _e , or sa		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		1363.5 cm ²	SG of Contents		0.681
Element Volume		9014.3 cm ³	Weight of Contents		6.1 Kg
Empty Weight		12.3 Kg	Operating Weight		18.5 Kg
			Comp. All. Ope. Stress		145.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.027 \cdot 206.5) / (138.0 \cdot 1.0 - 0.6 \cdot 25.027)$$

$$= 3.7865 + 1.5000 = 5.2865 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.027 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.5) / (206.5 + 0.6 \cdot 10.5)$$

$$= 68 - 0 = 68 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (205.0 + 0.6 \cdot 12.0)$$

$$= 78 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.027 \cdot (206.5 + 0.6 \cdot 10.5)) / (1.0 \cdot 10.5)$$

$$= 50.725 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.844 %

Minimum Design Metal Temperature Results:



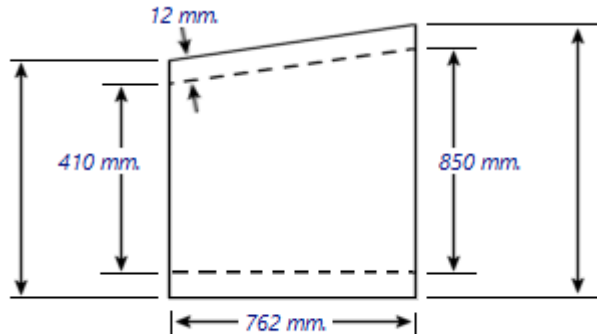
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL-2

Conical Head/Section Design Information:					
Design Pressure	P	25.057 bars	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		762.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	850.000 mm.	Actual Stress	119.1 N./mm ²	
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.	12.000 mm.	
Surface Area		16388.0 cm ²	SG of Contents	0.681	
Element Volume		215301.5 cm ³	Weight of Contents	146.5 Kg _f	
Empty Weight		165.5 Kg _f	Operating Weight	312.1 Kg _f	
			Comp. All. Ope. Stress	147.1 N./mm ²	
			To End Diameter	410.000 mm.	
Line of Support	Both Ends				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D) / (2 \cdot \cos(a) \cdot (S \cdot E - 0.6 \cdot P)) \text{ per Appendix 1-4 (e)}$$

$$= (25.057 \cdot 853.4642) / (2 \cdot 0.866 \cdot (138.0 \cdot 1.0 - 0.6 \cdot 25.057))$$

$$= 9.0460 + 1.5000 = 10.5460 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.057 bars

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a)) \text{ per App 1-4(e)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 10.5 \cdot 0.866) / (853.4642 + 1.2 \cdot 10.5 \cdot 0.866)$$

$$= 29 - 0.1 = 29 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a)) \text{ per App 1-4(e)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0 \cdot 0.866) / (850.0 + 1.2 \cdot 12.0 \cdot 0.866)$$

$$= 33 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (D + 1.2 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (25.057 \cdot (853.4642 + 1.2 \cdot 10.5 \cdot 0.866)) / (2 \cdot 1.0 \cdot 10.5 \cdot 0.866)$$

$$= 119.098 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.831 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: SHELL-2

Cone From 60 to 70 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.093	877.71	570.19	96.52	0.6496	0.0022663	106.48

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 106.4773) / (3 \cdot 96.5234) = 14.7075 \text{ bars}$$

Note:

The cone thickness used in the calculation has been modified per UG-33(f), $t_e = t \cdot \cos(\alpha)$.

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.348	877.71	570.19	373.79	0.6496	0.0002900	29.00

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 28.9958) / (3 \cdot 373.7888) = 1.0342 \text{ bars}$$

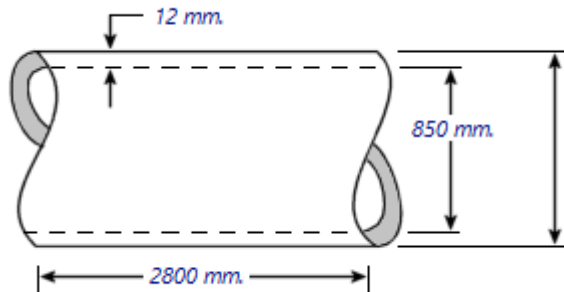
Note:

The cone thickness used in the calculation has been modified per UG-33(f), $t_e = t \cdot \cos(\alpha)$.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: K-SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	25.057 bars	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		2800.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	850.000 mm.	Actual Stress		103.3 N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000 mm.	Spec. Nominal Thk.		14.000 mm.
Surface Area		77232.9 cm ²	SG of Contents		0.681
Element Volume		1467865.0 cm ³	Weight of Contents		999.0 Kg
Empty Weight		824.6 Kg	Operating Weight		1823.6 Kg
			Comp. All. Ope. Stress		134.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.057 \cdot 426.5) / (138.0 \cdot 1.0 - 0.6 \cdot 25.057)$$

$$= 7.8298 + 1.5000 = 9.3298 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.057 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 10.5) / (426.5 + 0.6 \cdot 10.5)$$

$$= 33 - 0.1 = 33 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (425.0 + 0.6 \cdot 12.0)$$

$$= 38 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.057 \cdot (426.5 + 0.6 \cdot 10.5)) / (1.0 \cdot 10.5)$$

$$= 103.288 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.620 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: K-SHELL

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	874.00	2921.63	83.24	3.3428	0.0004881	48.80

$$MAEP = (4*B)/(3*(Do/t)) = (4*48.7973)/(3*83.2381) = 7.8161 \text{ bars}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.629	874.00	2921.63	188.79	3.3428	0.0001465	14.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.6451)/(3*188.7901) = 1.0343 \text{ bars}$$

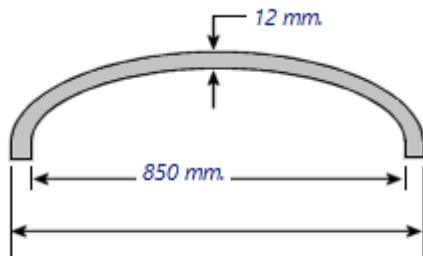
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.500	874.00	84181.38	83.24	50.0000	0.0001604	16.04

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.0374)/(3*83.2381) = 2.5688 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SH. COVER

Elliptical Head Design Information:							
Design Pressure	P	25.057	bars	Int. Design Temperature		85	°C
Material		SA-516	70	Ext. Design Temperature		85	°C
Length		50.800	mm.	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.	c	1.5000	mm.	Allowable Stress (ope)	S	138.0	N./mm ²
Ext. Corr. All.	ce	0.0000	mm.	Allowable Stress (amb)	Sa	138.0	N./mm ²
Inside Diameter	Di	850.000	mm.	Actual Stress		101.6	N./mm ²
Spec. Min. Thk t _e , or s _a		12.0000	mm.	Spec. Nominal Thk.		14.000	mm.
Surface Area		9757.5	cm ²	SG of Contents		0.681	
Element Volume		109234.8	cm ³	Weight of Contents		74.3	Kgf
Empty Weight		111.9	Kgf	Operating Weight		186.2	Kgf
				Comp. All. Ope. Stress		134.3	N./mm ²
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.625 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.313 mm.

Material treatment options: +N+IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.057 \cdot 853.0 \cdot 0.995) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.057)$$

$$= 7.7224 + 1.5000 = 9.2224 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.057 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 10.5) / (0.995 \cdot 853.0 + 0.2 \cdot 10.5)$$

$$= 34 - 0.1 = 34 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 850.0 + 0.2 \cdot 12.0)$$

$$= 39 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.057 \cdot (0.995 \cdot 853.0 + 0.2 \cdot 10.5)) / (2 \cdot 1.0 \cdot 10.5)$$



= 101.560 N./mm²

Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (25.057 \cdot 426.5) / (138.0 \cdot 1.0 - 0.6 \cdot 25.057) + 1.5 \\ &= 9.330 \text{ mm.} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.057 bars

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 12.5) / (426.5 + 0.6 \cdot 12.5) \\ &= 40 - 0.1 = 40 \text{ bars} \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6 \\ &= (2 + (853.0 / (2 \cdot 214.0)))^2 / 6 \\ &= 0.995335 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.931 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.625 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.313 mm.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: SH. COVER

Elliptical Head From 80 to 90 Ext. Chart: CS-2 at 85 °C

Elastic Modulus from Chart: CS-2 at 85 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
10.500	874.00	83.24	0.0016686	99.48

$$MAEP = B / (K_0 \cdot Do / t) = 99.4832 / (0.9 \cdot 83.2381) = 13.2788 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.263	874.00	386.22	0.0003596	35.95

$$MAEP = B / (K_0 \cdot Do / t) = 35.9527 / (0.9 \cdot 386.2236) = 1.0342 \text{ bars}$$

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	37.670 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	32.500 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	43.148 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	31.874 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	35.750 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	35.750 bars
App 27-4	= M.A.W.P.	28.977 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	43.807 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	32.500 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	43.776 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	37.067 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	35.750 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	35.750 bars
App 27-4	= M.A.W.P.	33.697 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
CH. COVER	85.8	234.0	0.367	43.85
CH. SHELL	88.9	234.0	0.380	43.85
SHELL-1	76.4	234.0	0.327	37.71
SHELL-2	179.4	234.0	0.767	37.75
K-SHELL	155.6	234.0	0.665	37.75
SH. COVER	153.0	234.0	0.654	37.75

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
TS IN NOZ. A1	Nozzle	118.00	118.00	1.000
TS IN NOZ. A1	Pad	138.00	138.00	1.000
TS OUT NOZ. B1	Nozzle	118.00	118.00	1.000
TS OUT NOZ. B1	Pad	138.00	138.00	1.000
SS IN NOZ A2	Nozzle	138.00	138.00	1.000
SS OUT NOZ B2	Nozzle	138.00	138.00	1.000
SS DRAIN NOZ D1	Nozzle	138.00	138.00	1.000
TEMP INST. T1	Nozzle	138.00	138.00	1.000
LEVEL INST L1	Nozzle	138.00	138.00	1.000
LEVEL INST L2	Nozzle	138.00	138.00	1.000
LEVEL INST L3	Nozzle	138.00	138.00	1.000
LEVEL INST L4	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH. COVER	138.00	138.00	1.000
CH. SHELL	138.00	138.00	1.000
CH. FLG	138.00	138.00	1.000
SH. FLG	138.00	138.00	1.000
SHELL-1	138.00	138.00	1.000
SHELL-2	138.00	138.00	1.000



K-SHELL	138.00	138.00	1.000
SH. COVER	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	118.00	118.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
TS IN NOZ. A1	18.50	216.00	0.086
TS OUT NOZ. B1	16.52	216.00	0.076
SS IN NOZ A2	17.79	179.40	0.099
SS OUT NOZ B2	37.02	179.40	0.206
SS DRAIN NOZ D1	14.22	179.40	0.079
TEMP INST. T1	14.22	179.40	0.079
LEVEL INST L1	14.65	179.40	0.082
LEVEL INST L2	14.65	179.40	0.082
LEVEL INST L3	14.65	179.40	0.082
LEVEL INST L4	14.65	179.40	0.082

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: TS

Shell Data:

Main Shell Description: SHELL-1

Shell Test Pressure		37.67 bars
Shell Thickness	ts	12.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	410.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Test Pressure		43.81 bars
Channel Thickness	tc	12.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	410.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	147
Tube Wall Thickness	et	2.1100 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	3600.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		3545.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	240.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	178.962 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	55.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	538.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	442.00 mm.
Step 1 Depth on the Tubeside	ht1	6.00 mm.
Step 2 Diameter on the Tubeside	dt2	407.00 mm.
Step 2 Depth on the Tubeside	ht2	6.00 mm.
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	442.00 mm.
Step 1 Depth on the Shellside	hs1	6.00 mm.
Step 2 Diameter on the Shellside	ds2	407.00 mm.
Step 2 Depth on the Shellside	hs2	6.00 mm.
Calculated Tubesheet Diameter per Div 2, 4.18.4		442.00 mm.

Note: Tubesheet diameter is now: 442.000 mm. Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	426.151 mm.
Area of the Untubed Lanes	AL	273.2 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	538.000 mm.
Flange Inside Diameter	B	410.000 mm.
Flange Face Outside Diameter	Fod	445.000 mm.
Flange Face Inside Diameter	Fid	410.000 mm.
Gasket Outside Diameter	Go	439.000 mm.
Gasket Inside Diameter	Gi	413.000 mm.
Small end Hub thk.	g0	10.0000 mm.
Large end Hub thk.	g1	13.0000 mm.
Gasket Factor,	m	2.00
Gasket Design Seating Stress	y	17.24 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm.
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	493.000 mm.
Nominal Bolt Diameter	dB	20.0000 mm.
Thread Series	TEMA Metric Thread	
Number of Bolts	n	16
Bolt Material		SA-320 L7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm.
Alternate Flange Operating Bolt Load, Wm1		79081.91 Kg
Alternate Flange Seating Bolt Load, Wm2		15118.27 Kg
Alternate Flange Design Bolt Load, W		60907.91 Kg

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with		None
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	31.0000 mm.
Is Bolt Load Transferred to the Tubesheet		Yes

Diagram illustrating Configuration d: Tubesheet Gasketed w/Shell and Channel. The diagram shows a central tubesheet (G_c) with 12 tubes. The tubesheet is gasketed to a shell (G_s) on the left and a channel (G_s) on the right. Dimensions include a total height of 493mm, a shell height of 538mm, a channel height of 538mm, a tube pitch of 43.8bars, a tube diameter of 410mm, a tube wall thickness of 37.7bars, and a tube length of 55mm.

TCA	ODCA	SLen	D/T	L/D	Factor A	B
2.1100	19.05	3723.96	9.03	50.0000	0.0134948	122.73

EMAWP = (2.167/(D/T)-0.0833)*B = 192.3322 bars



Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6072	19.05	3723.96	31.37	50.0000	0.0011177	88.65
$EMAWP = (4*B)/(3*(D/T)) = (4 * 88.647)/(3 * 31.3717) = 37.6738 \text{ bars}$						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6072 mm.
Allowable Internal Pressure at Corroded thickness	286.79 bars
Required Internal Design Pressure	43.81 bars
Allowable External Pressure at Corroded thickness	192.33 bars
Required External Design Pressure	37.67 bars
Required Thickness due to Shell Side pressure	0.6072 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm.
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm.
Effective Gasket Width,	$b = \sqrt{b_0} * 2.5$	6.422 mm.
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	426.156 mm.

Warning: Available bolt area, 34.728 cm² for then shell flange is < required 38.774 cm²

Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 * r_o + dt \\ &= 2 * 178.962 + 19.05 \\ &= 376.974 \text{ mm.} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (25.4 - 19.05) / 25.4 \\ &= 0.250 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= G_s / D_o \text{ (Configurations d, e, f)} \\ &= 426.1556 / 376.974 \\ &= 1.1305 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= G_c / D_o \text{ (Configurations d)} \\ &= 426.1509 / 376.974 \\ &= 1.1305 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 55.0 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2t_t (E_t T / E) * (S_t T / S) * (\rho); dt - 2t_t) \\ &= \max(19.05 - 2 * 2.11 (0.20270E+09 / .20270E+09) * \end{aligned}$$



$$\begin{aligned} & (117/138) \cdot (0); 19.05 - 2 \cdot 2.11 \\ & = 19.0500 \text{ mm.} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} & = p / \sqrt{(1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))} \\ & = 25.4 / \sqrt{(1 - 4 \cdot \min(273.18; 4 \cdot 376.974 \cdot 25.4) / (\pi \cdot 376.974^2))} \\ & = 29.2274 \text{ mm.} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} & = (p^* - d^*) / p^* \\ & = (29.2274 - 19.05) / 29.2274 \\ & = 0.3482 \end{aligned}$$

$$\begin{aligned} h/p & = 2.165354; \mu^* = 0.348215; E^* = 69892688 \text{ KPa.} \\ E^*/E & = 0.344806; \nu^* = 0.327284 \end{aligned}$$

Note: As h/p (2.165) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} & = A / Do \\ & = 442.0/376.974 \\ & = 1.1725 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} & = (1 - \nu^*) / E^* \cdot (E \cdot \ln(K)) \\ & = (1 - 0.33) / 69892688 \cdot (0.20270E+09 \cdot \ln(1.17)) \\ & = 0.3105 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} & = Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\ & = 376.974^2/16 \cdot [(1.13 - 1) \cdot (1.13^2 + 1) \cdot 37.67 - \\ & \quad (1.13 - 1) \cdot (1.13^2 + 1) \cdot 43.807] \\ & = -16186.2510 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} & = Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do) \\ & = -16186.3 + 79081.9 \cdot (426.151 - 426.151) / (2 \cdot \pi \cdot 376.974) \\ & = -16186.2510 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} & = (M^* - Do^2/32 \cdot F(Ps - Pt)) / (1 + F) \\ & = (-16186.25 - 376.974^2/32 \cdot 0.31(37.67 - 43.81)) / (1 + 0.31) \\ & = -5894.8188 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} & = Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt) \\ & = -5894.82 + 376.974^2/64 \cdot (3 + 0.327) \cdot (37.67 - 43.81) \\ & = -51234.3477 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} & = \max(|Mp|; |Mo|) \\ & = \max(|-5894.819|; |-51234.348|) \\ & = 51234.3477 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} & = 6 \cdot M / (\mu^* (h - hg')^2) \\ & = 6 \cdot 51234.3477 / (0.3482 (55.0 - 6.0)^2) \\ & = 36.7704 \text{ N/mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot \beta T \cdot S_y \\ &= 2 \cdot 0.95 \cdot 250.0 \\ &= 474.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 50897.625 / (0.3482 (19.6027 - 6.0)^2) \\ &= 474.0002 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 19.6027 + 0 + 0 \\ &= 19.6027 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(19.6027 ; 1.2177) \\ &= 19.6027 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |37.67 - 43.81| \\ &= 6.137 \text{ bars} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y \\ &= 4 \cdot 0.25 \cdot 55.0 / 376.97 \cdot 0.8 \cdot 0.95 \cdot 249.995 \\ &= 27.720 \text{ N./mm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta T \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.25) \cdot (376.97 / 189.997) \cdot |37.67 - 43.81| + 0 \\ &= 1.2177 \text{ mm.} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.11) 2.11 \\ &= 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.11 (19.05 - 2.11) 227.996 \\ &= 2610.623 \text{ Kg} \end{aligned}$$

Groove Weld Strength [F_g]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 228.0 \\ &= 2610.6226 \text{ Kg} \end{aligned}$$

Weld Strength Factor [f_w]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 227.996 / 227.996 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [F_f]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$



$$= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 228.0$$

$$= 1734.5131 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do})$$

$$= 2(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) \cdot 1.0 \cdot 1.0) - 0.75 \cdot 19.05})$$

$$= 2.562 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm.

Minimum Required Groove Weld Leg [agr]: 1.2810 mm.

Tube-Tubesheet Jt allowable, 2610.62 is >= tube strength 2610.62 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	36.8 <=	475.0 N./mm ²	Ok
Tubesheet shear stress	4.2 <=	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	19.603	55.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	42.450 ...	263	475 30 190	Ps+Pt D1	Ok
D2uc	39.809 20.456	226	475 26 190	Ps+Pt D2	Ok
D3uc	19.603 ...	37	475 4 190	Ps+Pt D3	Ok
Max:	42.4504 20.456 mm.	0.553	0.158 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
-----------------	----------	--------	-----

Tubesheet thickness :	42.450	55.000	Ok
Tubesheet thickness flanged extension :	20.456	31.000	Ok
Tube thickness :	0.607	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	434	10.5	0.0033602	113.835
20	30	384.167	434	10.5	0.0059037	121.159
30	40	No Calc	...	53.5	No Calc	No Calc
40	50	No Calc	...	53.5	No Calc	No Calc
50	60	100	434	10.5	0.030251	122.731
60	70	570.192	877.713	10.5	0.0022663	106.477
70	80	2921.63	874	10.5	0.00048808	48.7973
80	90	No Calc	874	10.5	0.0016686	99.4832

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	12	No Calc	...	30.5991
20	30	12	No Calc	...	39.0812
30	40	55	47.6504	...	No Calc
40	50	55	47.6504	...	No Calc
50	60	12	No Calc	...	39.5883
60	70	12	4.21141	1.03419	14.7075
70	80	12	6.12948	1.03419	7.81605
80	90	12	3.76294	1.03419	13.2788

Minimum

8

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	384.167	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	100	No Calc	No Calc	No Calc
60	70	570.192	570.192	No Calc	No Calc
70	80	2921.63	84181.4	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	30.5721	15625.8	27.2966	15922.3	3.05721
20	30	36.989	39614.7	32.4804	40196.6	3.6989
30	40	46.0869	10020.1	44.9598	10049.1	4.60869
40	50	46.0869	10020.1	44.9598	10049.1	4.60869
50	60	12.3297	9014.32	10.8268	9208.27	1.23297
60	70	165.521	215302	145.22	217921	16.5521
70	80	824.626	1467865	737.552	1479102	82.4626
80	90	111.901	109235	99.9118	110293	11.1901
Total		1274	1876696.25	1143	1892742.12	127



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	10.5566	-34.1667	...	...	Liquid 10
10	Insl	1.10058	-26.25	...	...	Ins: 10
20	Liqd	26.7632	150	...	...	Liquid: 20
20	Insl	1.33158	150	...	...	Ins: 20
20	Nozl	16.1461	150	238.325	...	TS IN NOZ. A1
20	Nozl	9.80626	150	-226.425	...	TS OUT NOZ. B1
30	Liqd	10.0139	35	...	...	Liquid 30
30	Insl	0.42302	35	...	...	Ins: 30
40	Liqd	10.0139	35	...	...	Liquid 40
40	Insl	0.42302	35	...	...	Ins: 40
50	Liqd	6.135	50	...	...	Liquid 50
50	Insl	0.44386	50	...	...	Ins: 50
60	Liqd	146.53	331.791	-110	...	Liquid 60
60	Insl	5.4638	381	...	...	Ins: 60
70	Sadl	87.4111	150	550.5	...	SLIDING SADDLE
70	Sadl	87.4111	2240	550.5	...	FIXED SADDLE
70	Liqd	999.006	1400	...	...	Liquid 70
70	Insl	23.278	1400	...	...	Ins: 70
70	Nozl	18.6391	2500	463.1	...	SS IN NOZ A2
70	Nozl	69.5623	1400	526.6	...	SS OUT NOZ B2
70	Nozl	10.9669	1400	-446.425	...	SS DRAIN NOZ D1
70	Nozl	10.9669	2600	-446.425	...	TEMP INST. T1
80	Liqd	74.3435	121.633	0.36335E-04	...	Liquid: 80
80	Insl	3.69383	131.65	...	...	Ins: 80
80	Nozl	8.61733	209.407	200	-200	LEVEL INST L1
80	Nozl	8.61733	209.407	-200	-200	LEVEL INST L2
80	Nozl	8.61733	209.407	200	200	LEVEL INST L3
80	Nozl	8.61733	209.407	-200	200	LEVEL INST L4
30	FTsh	86.9448	100.675	...	...	TS
30	Tube	534.147	1873.17	...	...	Tubes
40	Bafl	3.84433	710	...	...	Baffles
40	Bafl	3.84433	1420	...	...	Baffles
40	Bafl	3.84433	2130	...	...	Baffles
40	Bafl	3.84433	2840	...	...	Baffles
40	Bafl	3.84433	3550	...	...	Baffles

Total Weight of Each Detail Type:

Saddles 174.8
 Liquid 1283.4



Insulation	36.2
Nozzles	170.6
Exchanger Components	640.3
Liquid in Tubes	66.6

Sum of the Detail Weights 2371.8 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1401.5	1401.5	1401.5	1401.5	1401.5	1401.5
Saddles	174.8	174.8	174.8	174.8	174.8	174.8
Nozzles	170.6	170.6	170.6	170.6	170.6	170.6
Exchanger	640.3	640.3	640.3	640.3	640.3	640.3
Insulation	...	...	36.2	36.2	36.2	36.2
Ope. Liquid	...	...	...	...	...	1283.4
Tube Ope Lqd	...	...	...	...	...	66.6
Test Liquid	...	1875.6	...	...	...	...
Tube Tst Lqd	...	98.5	...	...	...	...
Totals	2387.2	4361.3	2423.4	2423.4	2423.4	3773.3

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	2387.2 kg.
Shop Test Wt.	- Fabricated weight + water (full)	4361.3 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	2423.4 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	2423.4 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	2423.4 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	3773.3 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	3629.3 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	98565.2 cm ³
Weight of Ope Liq in tubes :	66.6 kg.
Weight of Water in tubes :	98.5 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	2767.57	2466.21
20	30	4090.35	3864.16
30	40	2172.6	966.04
40	50	2172.6	966.04
50	60	1363.45	1288.05
60	70	16388	15697.5
70	80	77232.9	74769.9
80	90	9757.5	9188.33

Total	115945.0	109206.3 cm ²
Total	124.8	117.5 ft ²



Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
TS IN NOZ. A1	95.26	102.00	85	600	GR 1.1	...	...	...	...
TS OUT NOZ. B1	95.26	102.00	85	600	GR 1.1	...	...	...	...
SS IN NOZ A2	95.26	102.00	85	600	GR 1.1	...	...	...	...
SS OUT NOZ B2	47.63	51.00	85	300	GR 1.1	...	...	...	...
SS DRAIN NOZ D	47.63	51.00	85	300	GR 1.1	...	...	...	...
TEMP INST. T1	47.63	51.00	85	300	GR 1.1	...	...	...	...
LEVEL INST L1	47.63	51.00	85	300	GR 1.1	...	...	...	...
LEVEL INST L2	47.63	51.00	85	300	GR 1.1	...	...	...	...
LEVEL INST L3	47.63	51.00	85	300	GR 1.1	...	...	...	...
LEVEL INST L4	47.63	51.00	85	300	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 47.631 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 51.000 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 95.261 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 102.000 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.194 \cdot 0.989) / (1 + 1.7 \cdot 3.4 \cdot 0.194)))$
 $= \min(0.85, 0.919)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.768
Structure Height to Diameter ratio	5.104

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Z_g = 589788$ mm.

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 700.0 + 300.0 = 1000.0$ mm.
 $= 3.281$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because z (3.281 ft.) < 15 ft.
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1$, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 $K_1 = 0$, $K_2 = 0$, $K_3 = 0$



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.768 \cdot 1.266$$

$$= 27.4 \text{ lbs. } [12.4 \text{ Kg}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
CH. COVER	1000.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. SHELL	1000.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. FLG	1000.0	0.000	0.000	0.000	1.035	1.000	161.849
SH. FLG	1000.0	0.000	0.000	0.000	1.035	1.000	161.849
SHELL-1	1000.0	0.000	0.000	0.000	1.035	1.000	161.849
SHELL-2	1000.0	0.000	0.000	0.000	1.035	1.000	161.849
K-SHELL	1000.0	0.000	0.000	0.000	1.035	1.000	161.849
SH. COVER	1000.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1000	688.8	1176.16	161.849	13.6767
20	30	1000	688.8	2066.4	161.849	24.0286
30	40	1000	660	462	161.849	5.37226
40	50	1000	660	462	161.849	5.37226
50	60	1000	688.8	688.8	161.849	8.00955
60	70	1000	952.8	7260.34	161.849	84.4251
70	80	1000	1216.8	34070.4	161.849	396.18
80	90	1000	1216.8	2963.51	161.849	34.4605

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(3.69090.75)$
 = 0.053 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.053$, 0.03) per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.217 \cdot 3455 \cdot 1.5$
 = 337.851 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 3455$
 = 150.2 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	205	345.451	23.638
20	30	205	345.451	23.638
30	40	205	345.451	23.638
40	50	205	345.451	23.638
50	60	205	345.451	23.638
60	70	205	345.451	23.638



70	Sad1	425	345.451	49.0057
Sad1	80	425	345.451	49.0057
70	80	425	345.451	49.0057
80	90	425	345.451	49.0057

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	2605.175 mm.
Center of Gravity of the Liquid	2564.129 mm.
Center of Gravity of the Insulation	2432.695 mm.
Center of Gravity of the Nozzles	2935.635 mm.
Center of Gravity of the Tubesheet(s)	450.675 mm.
Center of Gravity of the Tubes	2223.175 mm.
Center of Gravity of the Baffles	2608.175 mm.
Center of Gravity of Bare Shell New and Cold	2383.339 mm.
Center of Gravity of Bare Shell Corroded	2375.198 mm.
Vessel CG in the Operating Condition	2409.960 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	2336.091 mm.
Vessel CG in the Test Condition	2440.005 mm.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

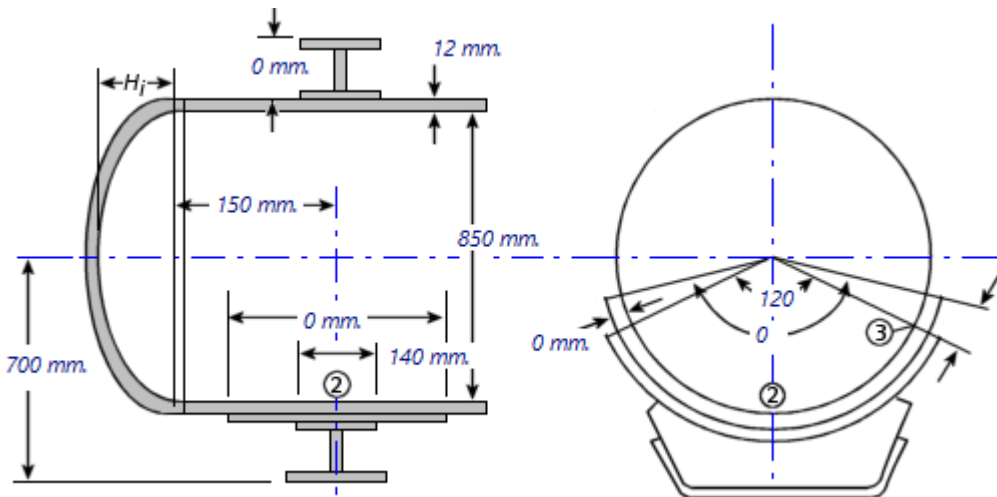
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{431.75 \cdot 10.5}, 2 \cdot 150.0)$$

$$= 245.0353 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	431.75	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	3782.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	150.00	mm.
Saddle Width	b1 or b	140.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	700.00	mm.
Coefficient of Friction	mu	0.10	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		2912.97	Kgf
Pressure used in Analysis	P	25.028	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	47.65	138.00
Long. Stress at Bottom of Midspan	55.27	138.00
Long. Stress at Top of Saddles	51.37	138.00
Long. Stress at Bottom of Saddles	51.55	138.00
Tangential Shear in Shell	5.54	110.40
Tangential Shear in Head	1.09	110.40
Circ. Stress at Horn of Saddle	7.91	172.50
Circ. Compressive Stress in Shell	0.84	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (571.5/2 + 0) \cdot 700.0/777.6909$
 $= 771.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(207.03, 0) \cdot 700.0/2090.0$
 $= 69.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(337.85, 0) \cdot 700.0/2090.0$
 $= 113.2 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (338/2 + 0) \cdot 700.0/777.6909$
 $= 456.1 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 2141 + \max(69, 772, 113, 456)$
 $= 2913.0 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169 (0.0 + \pi/4 (1.014 \cdot 1.2)^2)$
 $= 2030.240 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3000.38	Kgf
Transverse Shear Load	285.76	Kgf
Longitudinal Shear Load	337.85	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2 (ShellID/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2 (850.0/2 + 12.0 + 12.0) \sin(120.0/2), 2 \cdot 431.75)$
 $= 777.691 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -2913 \cdot 150.0 [1 - (1 - 150.0/3782.8 + (431.75^2 - 0.0^2)/(2 \cdot 150.0 \cdot 3782.8))/(1 + (4 \cdot 0.0)/(3 \cdot 3782.8))]$
 $= 54.4 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 2913 \cdot 3783/4 ((1 + 2(432^2 - 0^2)/(3783^2))/(1 + (4 \cdot 0)/(3 \cdot 3783)) - 4 \cdot 150/3783)$
 $= 2389.7 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 25.028 \cdot 431.75/(2 \cdot 10.5) - 2389.7/(\pi \cdot 431.8^2 \cdot 10.5)$
 $= 47.65 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 25.028 \cdot 431.75 / (2 \cdot 10.5) + 2389.7 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= 55.27 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 25.028 \cdot 431.75 / (2 \cdot 10.5) - 54.4 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= 51.37 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 25.028 \cdot 431.75 / (2 \cdot 10.5) + 54.4 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= 51.55 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 2913(3782.8 - 2 \cdot 150.0) / (3782.8 + (4 \cdot 0.0 / 3))$$

$$= 2682.0 \text{ Kg}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 2913 / (431.75 \cdot 10.5)$$

$$= 5.54 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 2913 / (431.75 \cdot 53.5)$$

$$= 1.09 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(431.75 \cdot 10.5)}$$

$$= 52.518 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 2913 \cdot 0.1 / (10.5(140.0 + 52.52 + 52.52))$$

$$= -0.84 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -2913 / (4 \cdot 10.5(140.0 + 52.518 + 52.518)) -$$

$$3 \cdot 0.0132 \cdot 2913 / (2 \cdot 10.5^2)$$

$$= -7.91 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{(431.75 \cdot 10.5)}, 2 \cdot 150.0)$$

$$= 245.04 \text{ mm.}$$

Distance between Saddle Supports [Ls]:

$$= 2090.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 2090.0(85.0 - 21.1)$$

$$= 1.603 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	820.0001	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	170.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		-32.0	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.



Friction Coefficient mu 0.100

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(426.5 \cdot 10.5)} + 180.0)^3) / 12$$

$$= 2.744 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 2.744 + 29.861 (109.326 - 5.25)^2$$

$$= 3237.299 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	284.4	10.5	5.2	29.9	15677.3	0.324E+04
Wearplate	180.0	12.0	16.5	21.6	35640.0	0.186E+04
Web	12.0	235.0	140.0	28.2	394800.0	0.156E+04
BasePlate	170.0	16.0	265.5	27.2	722159.9	0.664E+04
Totals	...	...	...	106.9	1168277.2	0.133E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1168277 / 106.861$$

$$= 109.326 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2912.972$$

$$= 592.8530 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 0.7551 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 700.0 - 426.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 347.287 \text{ mm.}$$

Bending Moment, M = Fh * d = 205.8944 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 1.6592 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(2913 + 87)170.0) / (2 \cdot 820.0 \cdot 174.663)}$$

$$= 7.238 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0)$$

$$= 777.691 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 777.6908 / (3 - 1)$$

$$= 388.845 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{pwid} / 2$$

$$= 388.8454 \cdot 170.0 / 2$$

$$= 330.519 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 3000.383 / (820.0 \cdot 170.0)$$

$$= 2.152 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 330.5 \cdot 2.15$$

$$= 711.393 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 15.36 + 23.331$$

$$= 38.691 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 711.4 / 38.6907$$

$$= 1.803 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	483.
Web	194.4	12.0	6.0	23.3	13998.4	183.
Totals	...	...	...	38.7	130734.4	666.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 140.0 - 12.0$$

$$= 128.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 130734.438 / 38.691$$

$$= 33.790 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(33.79, 140.0 - 33.79)$$

$$= 106.210 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{666.4 / 38.691}$$

$$= 41.500 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 700.0 - \cos(120.0/2) (425.0 + 12.0 + 12.0) - 16.0$$

$$= 459.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.20619\text{E}+09 / 262.0}$$

$$= 124.642$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 459.5/41.5$$

$$= 11.072$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 337.9 / (2 \cdot 820.0) \cdot 388.845 \cdot 459.5/2$$

$$= 18.404 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ (11.0722 < 124.6424) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.07)^2 / (2 \cdot 124.64^2)) 262 /$$

$$(5/3 + 3 \cdot (11.07) / (8 \cdot 124.64) - (11.07^3) / (8 \cdot 124.64^3))$$

$$= 153.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.8/153.52 + (18.4 \cdot 106.21/6663611) / 174.66$$

$$= 0.028$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	636.
Web	388.8	12.0	6.0	46.7	27996.9	146.
Totals	...	...	...	62.0	144732.9	782.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 144732.875/62.021$$

$$= 23.336 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(23.336, \text{abs}(140.0 - 23.336))$$

$$= 116.664 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 700.0 - \cos(449.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 235.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (781.6/62.021) }$$

$$= 35.498 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 235.0/35.498$$

$$= 6.620$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 337.851 / (2 \cdot 820.0)$$

$$= 0.206 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.206 \cdot 388.845 \cdot 235.0$$

$$= 18.825 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σB ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 18.825/66992.234$$

$$= 2.756 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (6.62 < 124.6424) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.62)^2 / (2 \cdot 124.64^2)) / 262 / (5/3 + 3 \cdot (6.62) / (8 \cdot 124.64) - (6.62^3) / (8 \cdot 124.64^3))$$

$$= 155.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 2.23/155.12 + (18.82 \cdot 116.664/781.558) / 174.66$$

$$= 0.030$$

Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{ 431.75 \cdot 10.5 }, 2 \cdot 560.0)$$

$$= 245.0353 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	431.75	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	3782.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	560.00	mm.
Saddle Width	b1 or b	140.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	219.25	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	700.00	mm.
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		2234.98	Kgf
Pressure used in Analysis	P	25.028	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		50.27	138.00
Long. Stress at Bottom of Midspan		52.65	138.00
Long. Stress at Top of Saddles		54.81	138.00
Long. Stress at Bottom of Saddles		49.60	138.00
Tangential Shear in Shell		3.70	110.40
Circ. Stress at Horn of Saddle		17.89	172.50
Circ. Compressive Stress in Shell		0.65	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(571.5/2 + 0) \cdot 700.0/777.6909$$

$$= 771.6 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(207.03, 0) \cdot 700.0/2090.0$$

= 69.3 Kgf

Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(337.85, 0) \cdot 700.0/2090.0$$

$$= 113.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(338/2 + 0) \cdot 700.0/777.6909$$

$$= 456.1 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 1463 + \max(69, 772, 113, 456)$$

$$= 2235.0 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169 (0.0 + \pi/4 (1.014 \cdot 1.2)^2)$$

$$= 2030.240 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2322.39	Kgf
Transverse Shear Load	285.76	Kgf
Longitudinal Shear Load	337.85	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(850.0/2 + 12.0 + 12.0) \sin(120.0/2), 2 \cdot 431.75)$$

$$= 777.691 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -2235 \cdot 560.0 [1 - (1 - 560.0/3782.8 + (431.75^2 - 219.25^2) / (2 \cdot 560.0 \cdot 3782.8)) / (1 + (4 \cdot 219.25) / (3 \cdot 3782.8))]$$

$$= -223.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 2235 \cdot 3783/4 ((1 + 2(432^2 - 219^2)/(3783^2)) / (1 + (4 \cdot 219)/(3 \cdot 3783)) - 4 \cdot 560/3783)$$

$$= 748.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 25.028 \cdot 431.75 / (2 \cdot 10.5) - 748.4 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= 50.27 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 25.028 \cdot 431.75 / (2 \cdot 10.5) + 748.4 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= 52.65 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 25.028 \cdot 431.75 / (2 \cdot 10.5) - 223.8 / (0.1066 \cdot \pi \cdot 431.8^2 \cdot 10.5)$$

$$= 54.81 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 25.028 \cdot 431.75 / (2 \cdot 10.5) + 223.8 / (0.1923 \cdot \pi \cdot 431.8^2 \cdot 10.5)$$

$$= 49.60 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 2235(3782.8 - 2 \cdot 560.0) / (3782.8 + (4 \cdot 219.25 / 3))$$

$$= 1460.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 1460.4 / (431.75 \cdot 10.5)$$

$$= 3.70 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(431.75 \cdot 10.5)}$$

$$= 52.518 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 2235 \cdot 0.1 / (10.5(140.0 + 52.52 + 52.52))$$

$$= -0.65 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -2235 / (4 \cdot 10.5(140.0 + 52.518 + 52.518)) - 3 \cdot 0.0529 \cdot 2235 / (2 \cdot 10.5^2)$$

$$= -17.89 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{(431.75 \cdot 10.5)}, 2 \cdot 560.0)$$

$$= 245.04 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	820.0001	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	170.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		-32.0	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(426.5 \cdot 10.5)} + 180.0)10.5^3) / 12$$

$$= 2.744 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 2.744 + 29.861(109.326 - 5.25)^2$$

$$= 3237.299 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	284.4	10.5	5.2	29.9	15677.3	0.324E+04
Wearplate	180.0	12.0	16.5	21.6	35640.0	0.186E+04
Web	12.0	235.0	140.0	28.2	394800.0	0.156E+04
BasePlate	170.0	16.0	265.5	27.2	722159.9	0.664E+04
Totals	...	...	...	106.9	1168277.2	0.133E+05

Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 1168277/106.861 \\ &= 109.326 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Angle } [\beta]: \\ &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120.0/2 \\ &= 120.0 \end{aligned}$$

$$\begin{aligned} \text{Saddle Splitting Coefficient [K1]:} \\ &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

$$\begin{aligned} \text{Saddle Splitting Force [Fh]:} \\ &= K1 \cdot Q \\ &= 0.204 \cdot 2234.981 \\ &= 454.8670 \text{ Kg} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh / As) = 0.5793 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Saddle Splitting Dimension [d]:} \\ &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 700.0 - 426.5 \cdot \sin(120.0/2) / 1.0472 \\ &= 347.287 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 157.9727 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.2730 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}) } \\ &= \sqrt{ 3(2235 + 87) 170.0 / (2 \cdot 820.0 \cdot 174.663) } \\ &= 6.368 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned} \text{Web Length Dimension [Web Length]:} \\ &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0) \\ &= 777.691 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Distance between Ribs [e]:} \\ &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 777.6908 / (3 - 1) \\ &= 388.845 \text{ mm.} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Pressure Area [Ap]:} \\ &= e \cdot \text{Bpwid} / 2 \\ &= 388.8454 \cdot 170.0/2 \\ &= 330.519 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Bearing Pressure [Bp]:} \\ &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 2322.392 / (820.0 \cdot 170.0) \\ &= 1.666 \text{ Kg/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Axial Load [P]:} \\ &= Ap \cdot Bp \\ &= 330.5 \cdot 1.67 \\ &= 550.641 \text{ Kg} \end{aligned}$$

$$\begin{aligned} \text{Area of the Rib and Web [Ar]:} \\ &= \text{Rib Area} + \text{Web Area} \\ &= 15.36 + 23.331 \\ &= 38.691 \text{ cm}^2 \end{aligned}$$

$$\text{Compressive Stress [Sc]:}$$

$$= P/Ar$$

$$= 550.6/38.6907$$

$$= 1.396 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	483.
Web	194.4	12.0	6.0	23.3	13998.4	183.
Totals	...	...	...	38.7	130734.4	666.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 140.0 - 12.0$$

$$= 128.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 130734.438/38.691$$

$$= 33.790 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(33.79, 140.0 - 33.79)$$

$$= 106.210 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(666.4/38.691)}$$

$$= 41.500 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 700.0 - \cos(120.0/2) (425.0 + 12.0 + 12.0) - 16.0$$

$$= 459.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.20619\text{E}+09/262.0)}$$

$$= 124.642$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 459.5/41.5$$

$$= 11.072$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 337.9/ (2 \cdot 820.0) \cdot 388.845 \cdot 459.5/2$$

$$= 18.404 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.0722 < 124.6424$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.07)^2 / (2 \cdot 124.64^2)) 262 / (5/3 + 3 \cdot (11.07) / (8 \cdot 124.64) - (11.07^3) / (8 \cdot 124.64^3))$$

$$= 153.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.4/153.52 + (18.4 \cdot 106.21/6663611) / 174.66$$

$$= 0.026$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



Rib	12.0	128.0	76.0	15.4	116736.0	636.
Web	388.8	12.0	6.0	46.7	27996.9	146.
Totals	...	...	...	62.0	144732.9	782.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 144732.875/62.021$$

$$= 23.336 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(23.336, \text{abs}(140.0 - 23.336))$$

$$= 116.664 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 700.0 - \cos(449.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 235.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (781.6/62.021) }$$

$$= 35.498 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 235.0/35.498$$

$$= 6.620$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 337.851/(2 \cdot 820.0)$$

$$= 0.206 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.206 \cdot 388.845 \cdot 235.0$$

$$= 18.825 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 18.825/66992.234$$

$$= 2.756 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.62 < 124.6424$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.62)^2 / (2 \cdot 124.64^2)) 262 / (5/3 + 3 \cdot (6.62) / (8 \cdot 124.64) - (6.62^3) / (8 \cdot 124.64^3))$$

$$= 155.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 1.73/155.12 + (18.82 \cdot 116.664/781.558) / 174.66$$

$$= 0.027$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	150.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	70.0 mm.
Nominal Bolt Diameter	Bnd	22.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	1.9179 cm ²
Saddle Load Q0 (Weight)	Q0	1550.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	113.2 Kg



Maximum Transverse Force	Ft	285.8 Kgf
Maximum Longitudinal Force	F1	337.9 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $337.9 / (1.918 \cdot 4)$
= 4.3 N./mm². Must be less than 100.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $285.8 / (1.918 \cdot 4)$
= 3.7 N./mm². Must be less than 100.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $337.85 / (100.0 \cdot 4.0)$
= 0.0828 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $700.0 \cdot 285.76 + 0.0$
= 200.04 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $200.04 / 1550.75$
= 128.99 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 0.0828, 0.0]$
= 0.0828 cm²

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

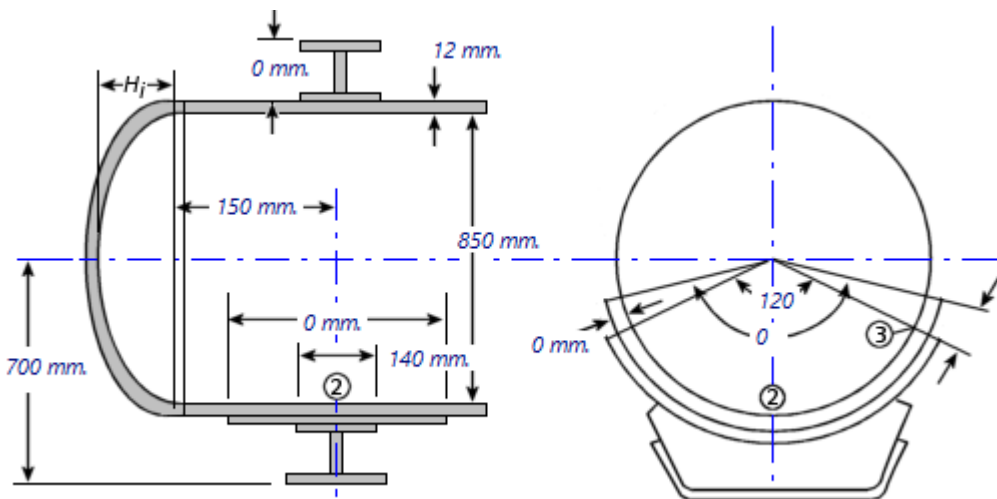
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{431.75 \cdot 10.5}, 2 \cdot 150.0)$$

$$= 245.0353 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	431.75	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	3782.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	150.00	mm.
Saddle Width	b1 or b	140.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	700.00	mm.
Coefficient of Friction	mu	0.10	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		2713.47	Kgf
Pressure used in Analysis	P	37.712	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	73.99	234.00
Long. Stress at Bottom of Midspan	81.09	234.00
Long. Stress at Top of Saddles	77.46	234.00
Long. Stress at Bottom of Saddles	77.62	234.00
Tangential Shear in Shell	5.16	138.58
Tangential Shear in Head	1.01	122.38
Circ. Stress at Horn of Saddle	7.37	350.99
Circ. Compressive Stress in Shell	0.79	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(188.6/2 + 0) \cdot 700.0/777.6909$
 $= 254.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(68.32, 0) \cdot 700.0/2090.0$
 $= 22.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 2459 + \max(23, 255, 0, 0)$
 $= 2713.5 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.014 \cdot 1.2)^2)$
 $= 2030.240 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2800.88	Kgf
Transverse Shear Load	94.30	Kgf
Longitudinal Shear Load	68.32	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(850.0/2 + 12.0 + 12.0)\sin(120.0/2), 2 \cdot 431.75)$
 $= 777.691 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a[1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -2713 \cdot 150.0[1 - (1 - 150.0/3782.8 + (431.75^2 - 0.0^2)/(2 \cdot 150.0 \cdot 3782.8))/(1 + (4 \cdot 0.0)/(3 \cdot 3782.8))]$
 $= 50.7 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 2713 \cdot 3783/4((1 + 2(432^2 - 0^2)/(3783^2))/(1 + (4 \cdot 0)/(3 \cdot 3783)) - 4 \cdot 150/3783)$
 $= 2226.0 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 37.712 \cdot 431.75/(2 \cdot 10.5) - 2226.0/(\pi \cdot 431.8^2 \cdot 10.5)$
 $= 73.99 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 37.712 \cdot 431.75/(2 \cdot 10.5) + 2226.0/(\pi \cdot 431.8^2 \cdot 10.5)$
 $= 81.09 \text{ N/mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:
 $= P \cdot R_m/(2t) - M_1/(\pi \cdot R_m^2 \cdot t)$
 $= 37.712 \cdot 431.75/(2 \cdot 10.5) - 50.7/(\pi \cdot 431.8^2 \cdot 10.5)$
 $= 77.46 \text{ N/mm}^2$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 t)$$

$$= 37.712 \cdot 431.75 / (2 \cdot 10.5) + 50.7 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= 77.62 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 2713(3782.8 - 2 \cdot 150.0) / (3782.8 + (4 \cdot 0.0 / 3))$$

$$= 2498.3 \text{ Kg}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 2713 / (431.75 \cdot 10.5)$$

$$= 5.16 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 2713 / (431.75 \cdot 53.5)$$

$$= 1.01 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(431.75 \cdot 10.5)}$$

$$= 52.518 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 2713 \cdot 0.1 / (10.5(140.0 + 52.52 + 52.52))$$

$$= -0.79 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -2713 / (4 \cdot 10.5(140.0 + 52.518 + 52.518)) -$$

$$3 \cdot 0.0132 \cdot 2713 / (2 \cdot 10.5^2)$$

$$= -7.37 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{(431.75 \cdot 10.5)}, 2 \cdot 150.0)$$

$$= 245.04 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	820.0001	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	170.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(426.5 \cdot 10.5)} + 180.0) 10.5^3) / 12$$

$$= 2.744 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 2.744 + 29.861(109.326 - 5.25)^2$$

$$= 3237.299 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	284.4	10.5	5.2	29.9	15677.3	0.324E+04
Wearplate	180.0	12.0	16.5	21.6	35640.0	0.186E+04



Web	12.0	235.0	140.0	28.2	394800.0	0.156E+04
BasePlate	170.0	16.0	265.5	27.2	722159.9	0.664E+04
Totals	...	...	...	106.9	1168277.2	0.133E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1168277/106.861$$

$$= 109.326 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2713.47$$

$$= 552.2500 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 0.7034 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 700.0 - 426.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 347.287 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 191.7932 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.5456 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}) }$$

$$= \sqrt{ 3(2713 + 87)170.0 / (2 \cdot 820.0 \cdot 173.33) }$$

$$= 7.020 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0)$$

$$= 777.691 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 777.6908 / (3 - 1)$$

$$= 388.845 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 388.8454 \cdot 170.0/2$$

$$= 330.519 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2800.881 / (820.0 \cdot 170.0)$$

$$= 2.009 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 330.5 \cdot 2.01$$

$$= 664.091 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 15.36 + 23.331 \\
 &= 38.691 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P/Ar \\
 &= 664.1/38.6907 \\
 &= 1.683 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	483.
Web	194.4	12.0	6.0	23.3	13998.4	183.
Totals	...	...	...	38.7	130734.4	666.

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 140.0 - 12.0 \\
 &= 128.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 130734.438/38.691 \\
 &= 33.790 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \max(\text{ytot}, \text{Saddle Width} - \text{ytot}) \\
 &= \max(33.79, 140.0 - 33.79) \\
 &= 106.210 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 666.4/38.691 } \\
 &= 41.500 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 700.0 - \cos(120.0/2) (425.0 + 12.0 + 12.0) - 16.0 \\
 &= 459.500 \text{ mm.}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ \frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}} } \\
 &= \sqrt{ \frac{2 \cdot \pi^2 \cdot 0.20619\text{E}+09}{260.0} } \\
 &= 125.121
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 459.5/41.5 \\
 &= 11.072
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2 \\
 &= 68.3/ (2 \cdot 820.0) \cdot 388.845 \cdot 459.5/2 \\
 &= 3.722 \text{ Kg-m.}
 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($11.0722 < 125.1208$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2/(2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r)/(8 \cdot Cc) - (KL/r)^3/(8 \cdot Cc^3)) \\
 &= (1 - (11.07)^2/(2 \cdot 125.12^2)) 260 / \\
 &\quad (5/3 + 3 \cdot (11.07)/(8 \cdot 125.12) - (11.07^3)/(8 \cdot 125.12^3)) \\
 &= 152.4 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 1.68/152.36 + (3.72 \cdot 106.21/6663611) / 173.33 \\
 &= 0.014
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	636.
Web	388.8	12.0	6.0	46.7	27996.9	146.
Totals	...	...	...	62.0	144732.9	782.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 144732.875/62.021$$

$$= 23.336 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(23.336, \text{abs}(140.0 - 23.336))$$

$$= 116.664 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 700.0 - \cos(449.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 235.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (781.6/62.021) }$$

$$= 35.498 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 235.0/35.498$$

$$= 6.620$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 68.32/(2 \cdot 820.0)$$

$$= 0.042 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.042 \cdot 388.845 \cdot 235.0$$

$$= 3.807 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 3.807/66992.234$$

$$= 0.557 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (6.62 < 125.1208) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.62)^2 / (2 \cdot 125.12^2)) 260 /$$

$$(5/3 + 3 \cdot (6.62) / (8 \cdot 125.12) - (6.62^3) / (8 \cdot 125.12^3))$$

$$= 153.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 2.08/153.95 + (3.81 \cdot 116.664/781.558) / 173.33$$

$$= 0.017$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ (Rm \cdot t) }, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{ (431.75 \cdot 10.5) }, 2 \cdot 560.0)$$

$$= 245.0353 \text{ mm.}$$



Input and Calculated Values:

Vessel Mean Radius	Rm	431.75	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	3782.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	560.00	mm.
Saddle Width	b1 or b	140.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	219.25	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	700.00	mm.
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		2018.42	Kgf
Pressure used in Analysis	P	37.712	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	76.46	234.00
Long. Stress at Bottom of Midspan	78.62	234.00
Long. Stress at Top of Saddles	80.56	234.00
Long. Stress at Bottom of Saddles	75.86	234.00
Tangential Shear in Shell	3.34	138.58
Circ. Stress at Horn of Saddle	16.16	350.99
Circ. Compressive Stress in Shell	0.58	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0(188.6/2 + 0) \cdot 700.0/777.6909$
 $= 254.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,\text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(68.32, 0) \cdot 700.0/2090.0$
 $= 22.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 1764 + \max(23, 255, 0, 0)$
 $= 2018.4 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587.169(0.0 + \pi/4(1.014 \cdot 1.2)^2)$
 $= 2030.240 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2105.83	Kgf
Transverse Shear Load	94.30	Kgf
Longitudinal Shear Load	68.32	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$

$$= \min(2(850.0/2 + 12.0 + 12.0) \sin(120.0/2), 2 \cdot 431.75)$$

$$= 777.691 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -2018 \cdot 560.0 [1 - (1 - 560.0/3782.8 + (431.75^2 - 219.25^2) / (2 \cdot 560.0 \cdot 3782.8)) / (1 + (4 \cdot 219.25) / (3 \cdot 3782.8))]$$

$$= -202.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 2018 \cdot 3783 / 4 \cdot ((1 + 2(432^2 - 219^2)/(3783^2)) / (1 + (4 \cdot 219) / (3 \cdot 3783)) - 4 \cdot 560 / 3783)$$

$$= 675.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 37.712 \cdot 431.75 / (2 \cdot 10.5) - 675.8 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= 76.46 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 37.712 \cdot 431.75 / (2 \cdot 10.5) + 675.8 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= 78.62 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 37.712 \cdot 431.75 / (2 \cdot 10.5) - 202.2 / (0.1066 \cdot \pi \cdot 431.8^2 \cdot 10.5)$$

$$= 80.56 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 37.712 \cdot 431.75 / (2 \cdot 10.5) + 202.2 / (0.1923 \cdot \pi \cdot 431.8^2 \cdot 10.5)$$

$$= 75.86 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm/3))$$

$$= 2018(3782.8 - 2 \cdot 560.0) / (3782.8 + (4 \cdot 219.25/3))$$

$$= 1318.9 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 1318.89 / (431.75 \cdot 10.5)$$

$$= 3.34 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(431.75 \cdot 10.5)}$$

$$= 52.518 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 2018 \cdot 0.1 / (10.5(140.0 + 52.52 + 52.52))$$

$$= -0.58 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -2018 / (4 \cdot 10.5(140.0 + 52.518 + 52.518)) - 3 \cdot 0.0529 \cdot 2018 / (2 \cdot 10.5^2)$$

$$= -16.16 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$



$$= \min(140.0 + 1.56 \cdot \sqrt{(431.75 \cdot 10.5)}, 2 \cdot 560.0)$$

$$= 245.04 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	820.0001	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	170.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(426.5 \cdot 10.5)} + 180.0) 10.5^3) / 12$$

$$= 2.744 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 2.744 + 29.861 (109.326 - 5.25)^2$$

$$= 3237.299 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	284.4	10.5	5.2	29.9	15677.3	0.324E+04
Wearplate	180.0	12.0	16.5	21.6	35640.0	0.186E+04
Web	12.0	235.0	140.0	28.2	394800.0	0.156E+04
BasePlate	170.0	16.0	265.5	27.2	722159.9	0.664E+04
Totals	...	...	...	106.9	1168277.2	0.133E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1168277 / 106.861$$

$$= 109.326 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2018.424$$

$$= 410.7930 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.5232 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 700.0 - 426.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 347.287 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 142.6660 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.1497 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$



Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \text{BasePlateWidth}}{(2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}}$$

$$= \sqrt{\frac{3(2018 + 87)170.0}{(2 \cdot 820.0 \cdot 173.33)}}$$

$$= 6.087 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0)$$

$$= 777.691 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 777.6908 / (3 - 1)$$

$$= 388.845 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 388.8454 \cdot 170.0/2$$

$$= 330.519 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2105.835 / (820.0 \cdot 170.0)$$

$$= 1.511 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 330.5 \cdot 1.51$$

$$= 499.295 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 15.36 + 23.331$$

$$= 38.691 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/\text{Ar}$$

$$= 499.3/38.6907$$

$$= 1.266 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	483.
Web	194.4	12.0	6.0	23.3	13998.4	183.
Totals	...	...	...	38.7	130734.4	666.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 140.0 - 12.0$$

$$= 128.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 130734.438/38.691$$

$$= 33.790 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(33.79, 140.0 - 33.79)$$

$$= 106.210 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\frac{\text{Total Inertia}}{\text{Total Area}}}$$

$$= \sqrt{666.4/38.691}$$

= 41.500 mm.

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 700.0 - cos(120.0/2)(425.0 + 12.0 + 12.0) - 16.0
 = 459.500 mm.

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.20619\text{E}+09/260.0)}$
 = 125.121

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 459.5/41.5
 = 11.072

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 68.3/ (2 • 820.0) • 388.845 • 459.5/2
 = 3.722 Kg-m.

Compressive Allowable, KL/r < Cc (11.0722 < 125.1208) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))
 = (1-(11.07)²/(2 • 125.12²))260/
 (5/3+3•(11.07)/(8• 125.12)-(11.07³)/(8•125.12³))
 = 152.4 N/mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 1.27/152.36 + (3.72 • 106.21/6663611)/173.33
 = 0.012

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	636.
Web	388.8	12.0	6.0	46.7	27996.9	146.
Totals	...	...	...	62.0	144732.9	782.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 144732.875/62.021
 = 23.336 mm.

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))
 = max(23.336, abs(140.0 - 23.336))
 = 116.664 mm.

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 700.0 - cos(449.0/2)(16.0 + 0.0 + 0.0) - 0.0
 = 235.000 mm.

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$
 = $\sqrt{(781.6/62.021)}$
 = 35.498 mm.

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 235.0/35.498
 = 6.620

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 68.32/ (2 • 820.0)

= 0.042 Kg/mm.

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 0.042 • 388.845 • 235.0
 = 3.807 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 3.807/66992.234
 = 0.557 N./mm²

Compressive Allowable, $KL/r < C_c$ (6.62 < 125.1208) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)
 = (1-(6.62)²/(2 • 125.12²))260/
 (5/3+3•(6.62)/(8• 125.12)-(6.62³)/(8•125.12³)
 = 153.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$
 = 1.57/153.95 + (3.81 • 116.664/781.558)/173.33
 = 0.013

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	70.0 mm.
Nominal Bolt Diameter	Bnd	22.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.9179 cm ²
Saddle Load Q0 (Weight)	Q0	1851.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	22.9 Kg
Maximum Transverse Force	Ft	94.3 Kg
Maximum Longitudinal Force	Fl	68.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 68.3/(1.918 • 4)
 = 0.9 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 94.3/(1.918 • 4)
 = 1.2 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarear!]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = 68.32/(103.2 • 4.0)
 = 0.0162 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$
 = 700.0 • 94.3 + 0.0
 = 66.01 Kg-m.

Eccentricity (e):

= R_{mom} / Q_0



= 66.01/1851.19
= 35.66 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.0162, 0.0]
= 0.0162 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

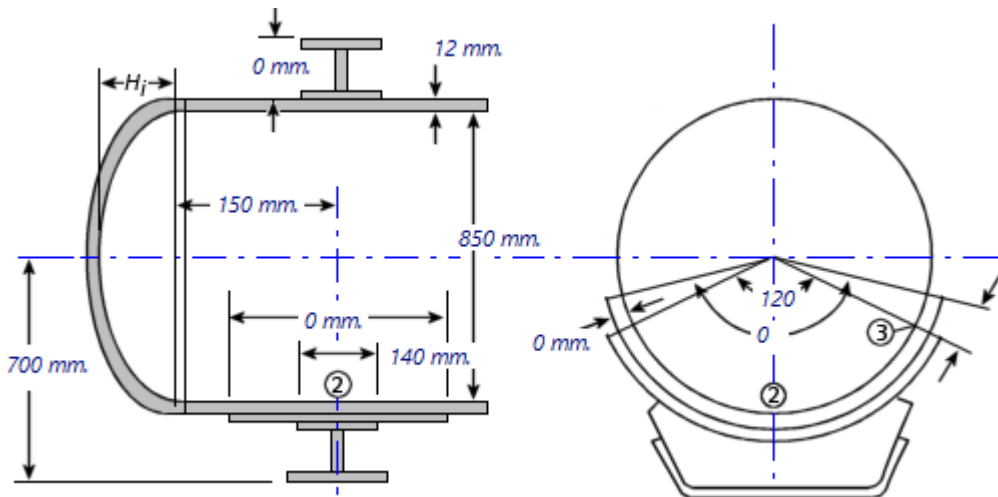
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{431.75 \cdot 10.5}, 2 \cdot 150.0)$$

$$= 245.0353 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	431.75	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	3782.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	150.00	mm.
Saddle Width	b1 or b	140.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	700.00	mm.
Coefficient of Friction	mu	0.10	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		2912.97	Kgf
Pressure used in Analysis	P	-1.034	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-5.94	-217.80
Long. Stress at Top of Midspan	5.94	138.00
Long. Stress at Bottom of Midspan	1.68	138.00
Long. Stress at Top of Saddles	-2.21	-217.80
Long. Stress at Top of Saddles	2.21	138.00
Long. Stress at Bottom of Saddles	-2.04	-217.80
Long. Stress at Bottom of Saddles	2.04	138.00
Tangential Shear in Shell	5.54	110.40



Tangential Shear in Head	1.09	110.40
Circ. Stress at Horn of Saddle	7.91	172.50
Circ. Compressive Stress in Shell	0.84	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(571.5/2 + 0) \cdot 700.0/777.6909$$

$$= 771.6 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(207.03, 0) \cdot 700.0/2090.0$$

$$= 69.3 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(337.85, 0) \cdot 700.0/2090.0$$

$$= 113.2 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(338/2 + 0) \cdot 700.0/777.6909$$

$$= 456.1 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2141 + \max(69, 772, 113, 456)$$

$$= 2913.0 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(1.014 \cdot 1.2)^2)$$

$$= 2030.240 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3000.38	Kgf
Transverse Shear Load	285.76	Kgf
Longitudinal Shear Load	337.85	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(850.0/2 + 12.0 + 12.0)\sin(120.0/2), 2 \cdot 431.75)$$

$$= 777.691 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is less than Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -2913 \cdot 150.0 [1 - (1 - 150.0/3782.8 + (431.75^2 - 0.0^2)/(2 \cdot 150.0 \cdot 3782.8))/(1 + (4 \cdot 0.0)/(3 \cdot 3782.8))]$$

$$= 54.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 2913 \cdot 3783/4((1 + 2(432^2 - 0^2)/(3783^2))/(1 + (4 \cdot 0)/(3 \cdot 3783)) - 4 \cdot 150/3783)$$

$$= 2389.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$



$$= -1.034 \cdot 431.75 / (2 \cdot 10.5) - 2389.7 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= -5.94 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.034 \cdot 431.75 / (2 \cdot 10.5) + 2389.7 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= 1.68 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.034 \cdot 431.75 / (2 \cdot 10.5) - 54.4 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= -2.21 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.034 \cdot 431.75 / (2 \cdot 10.5) + 54.4 / (\pi \cdot 431.8^2 \cdot 10.5)$$

$$= -2.04 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 2913(3782.8 - 2 \cdot 150.0) / (3782.8 + (4 \cdot 0.0 / 3))$$

$$= 2682.0 \text{ Kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 2913 / (431.75 \cdot 10.5)$$

$$= 5.54 \text{ N./mm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_3^*]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 2913 / (431.75 \cdot 53.5)$$

$$= 1.09 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(431.75 \cdot 10.5)}$$

$$= 52.518 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 2913 \cdot 0.1 / (10.5(140.0 + 52.52 + 52.52))$$

$$= -0.84 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -2913 / (4 \cdot 10.5(140.0 + 52.518 + 52.518)) - 3 \cdot 0.0132 \cdot 2913 / (2 \cdot 10.5^2)$$

$$= -7.91 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{(431.75 \cdot 10.5)}, 2 \cdot 150.0)$$

$$= 245.04 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	820.0001	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	170.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		-32.0	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(426.5 \cdot 10.5)} + 180.0) 10.5^3) / 12$$

$$= 2.744 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 2.744 + 29.861 (109.326 - 5.25)^2$$

$$= 3237.299 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	284.4	10.5	5.2	29.9	15677.3	0.324E+04
Wearplate	180.0	12.0	16.5	21.6	35640.0	0.186E+04
Web	12.0	235.0	140.0	28.2	394800.0	0.156E+04
BasePlate	170.0	16.0	265.5	27.2	722159.9	0.664E+04
Totals	...	...	...	106.9	1168277.2	0.133E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1168277 / 106.861$$

$$= 109.326 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2912.972$$

$$= 592.8530 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 0.7551 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 700.0 - 426.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 347.287 \text{ mm.}$$

Bending Moment, M = Fh * d = 205.8944 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 1.6592 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2913 + 87) 170.0 / (2 \cdot 820.0 \cdot 174.663)}$$

$$= 7.238 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0)$$

$$= 777.691 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 777.6908 / (3 - 1)$$

$$= 388.845 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 388.8454 \cdot 170.0/2$$

$$= 330.519 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3000.383 / (820.0 \cdot 170.0)$$

$$= 2.152 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 330.5 \cdot 2.15$$

$$= 711.393 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 15.36 + 23.331$$

$$= 38.691 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 711.4 / 38.6907$$

$$= 1.803 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	483.
Web	194.4	12.0	6.0	23.3	13998.4	183.
Totals	...	...	...	38.7	130734.4	666.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 140.0 - 12.0$$

$$= 128.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 130734.438 / 38.691$$

$$= 33.790 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(33.79, 140.0 - 33.79)$$

$$= 106.210 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(666.4 / 38.691)}$$

$$= 41.500 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 700.0 - \cos(120.0/2) (425.0 + 12.0 + 12.0) - 16.0$$

$$= 459.500 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.20619\text{E}+09 / 262.0)}$$

$$= 124.642$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 459.5 / 41.5$$

$$= 11.072$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$



$$= 337.9 / (2 \cdot 820.0) \cdot 388.845 \cdot 459.5 / 2$$

$$= 18.404 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ (11.0722 < 124.6424) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.07)^2 / (2 \cdot 124.64^2)) 262 /$$

$$(5/3 + 3 \cdot (11.07) / (8 \cdot 124.64) - (11.07^3) / (8 \cdot 124.64^3))$$

$$= 153.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.8 / 153.52 + (18.4 \cdot 106.21 / 6663611) / 174.66$$

$$= 0.028$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	636.
Web	388.8	12.0	6.0	46.7	27996.9	146.
Totals	...	...	...	62.0	144732.9	782.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 144732.875 / 62.021$$

$$= 23.336 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(23.336, \text{abs}(140.0 - 23.336))$$

$$= 116.664 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 700.0 - \cos(449.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 235.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{781.6 / 62.021}$$

$$= 35.498 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 235.0 / 35.498$$

$$= 6.620$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 337.851 / (2 \cdot 820.0)$$

$$= 0.206 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.206 \cdot 388.845 \cdot 235.0$$

$$= 18.825 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 18.825 / 66992.234$$

$$= 2.756 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (6.62 < 124.6424) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (6.62)^2 / (2 \cdot 124.64^2)) 262 /$$

$$(5/3 + 3 \cdot (6.62) / (8 \cdot 124.64) - (6.62^3) / (8 \cdot 124.64^3))$$

$$= 155.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)



$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$

$$= 2.23/155.12 + (18.82 \cdot 116.664/781.558)/174.66$$

$$= 0.030$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{ 431.75 \cdot 10.5 }, 2 \cdot 560.0)$$

$$= 245.0353 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	431.75	mm.
Shell Thickness used in this Case	t	12.000	mm.
Stiffened Vessel Length per 4.15.6	L	3782.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	560.00	mm.
Saddle Width	b1 or b	140.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	219.25	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	700.00	mm.
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		2234.98	Kgf
Pressure used in Analysis	P	-1.034	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-3.32	-217.80
Long. Stress at Top of Midspan	3.32	138.00
Long. Stress at Bottom of Midspan	-0.93	-217.80
Long. Stress at Bottom of Midspan	0.93	138.00
Long. Stress at Top of Saddles	1.22	138.00
Long. Stress at Bottom of Saddles	-3.98	-217.80
Long. Stress at Bottom of Saddles	3.98	138.00
Tangential Shear in Shell	3.70	110.40
Circ. Stress at Horn of Saddle	17.89	172.50
Circ. Compressive Stress in Shell	0.65	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(571.5/2 + 0) \cdot 700.0/777.6909$$

$$= 771.6 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(207.03, 0) \cdot 700.0/2090.0$$

$$= 69.3 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$



$$= \max(337.85, 0) \cdot 700.0/2090.0$$

$$= 113.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(338/2 + 0) \cdot 700.0/777.6909$$

$$= 456.1 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 1463 + \max(69, 772, 113, 456)$$

$$= 2235.0 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587.169(0.0 + \pi/4(1.014 \cdot 1.2)^2)$$

$$= 2030.240 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2322.39	Kgf
Transverse Shear Load	285.76	Kgf
Longitudinal Shear Load	337.85	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(850.0/2 + 12.0 + 12.0)\sin(120.0/2), 2 \cdot 431.75)$$

$$= 777.691 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -2235 \cdot 560.0 [1 - (1 - 560.0/3782.8 + (431.75^2 - 219.25^2)/(2 \cdot 560.0 \cdot 3782.8))/(1 + (4 \cdot 219.25)/(3 \cdot 3782.8))]$$

$$= -223.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 2235 \cdot 3783/4((1 + 2(432^2 - 219^2)/(3783^2))/(1 + (4 \cdot 219)/(3 \cdot 3783)) - 4 \cdot 560/3783)$$

$$= 748.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= -1.034 \cdot 431.75/(2 \cdot 10.5) - 748.4/(\pi \cdot 431.8^2 \cdot 10.5)$$

$$= -3.32 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= -1.034 \cdot 431.75/(2 \cdot 10.5) + 748.4/(\pi \cdot 431.8^2 \cdot 10.5)$$

$$= -0.93 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= -1.034 \cdot 431.75/(2 \cdot 10.5) - 223.8/(0.1066 \cdot \pi \cdot 431.8^2 \cdot 10.5)$$

$$= 1.22 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

$$= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.034 \cdot 431.75/(2 \cdot 10.5) + 223.8/(0.1923 \cdot \pi \cdot 431.8^2 \cdot 10.5)$$

$$= -3.98 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 2235(3782.8 - 2 \cdot 560.0)/(3782.8 + (4 \cdot 219.25/3))$$

$$= 1460.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 1460.4 / (431.75 \cdot 10.5)$$

$$= 3.70 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{431.75 \cdot 10.5}$$

$$= 52.518 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 2235 \cdot 0.1 / (10.5(140.0 + 52.52 + 52.52))$$

$$= -0.65 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q/(4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q/(2 \cdot t^2)$$

$$= -2235/(4 \cdot 10.5(140.0 + 52.518 + 52.518)) - 3 \cdot 0.0529 \cdot 2235/(2 \cdot 10.5^2)$$

$$= -17.89 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(140.0 + 1.56 \cdot \sqrt{431.75 \cdot 10.5}, 2 \cdot 560.0)$$

$$= 245.04 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	820.0001	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	170.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm.
Web Thickness	Webtk	12.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		-32.0	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})^3)/12$$

$$= ((1.56 \cdot \sqrt{426.5 \cdot 10.5} + 180.0)^3)/12$$

$$= 2.744 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 2.744 + 29.861(109.326 - 5.25)^2$$

$$= 3237.299 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	284.4	10.5	5.2	29.9	15677.3	0.324E+04
Wearplate	180.0	12.0	16.5	21.6	35640.0	0.186E+04
Web	12.0	235.0	140.0	28.2	394800.0	0.156E+04
BasePlate	170.0	16.0	265.5	27.2	722159.9	0.664E+04
Totals	...	...	...	106.9	1168277.2	0.133E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1168277/106.861$$

$$= 109.326 \text{ mm.}$$

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120.0/2
 = 120.0

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.204 • 2234.981
 = 454.8670 Kgf

Tension Stress, St = (Fh/As) = 0.5793 N./mm²
 Allowed Stress, Sa = 0.6 • Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 700.0 - 426.5 \cdot \sin(120.0/2) / 1.0472$$

 = 347.287 mm.

Bending Moment, M = Fh • d = 157.9727 Kg-m.

Bending Stress, Sb = (M • C1 / I) = 1.2730 N./mm²
 Allowed Stress, Sa = 2/3 • Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2235 + 87)170.0 / (2 \cdot 820.0 \cdot 174.663)}$$

 = 6.368 mm.

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (425.0 + 12.0 + 12.0)$$

 = 777.691 mm.

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 777.6908 / (3 - 1)$$

 = 388.845 mm.

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 388.8454 \cdot 170.0 / 2$$

 = 330.519 cm²

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2322.392 / (820.0 \cdot 170.0)$$

 = 1.666 Kgf/cm²

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 330.5 \cdot 1.67$$

 = 550.641 Kgf

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 15.36 + 23.331$$

 = 38.691 cm²

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 550.6 / 38.6907$$

 = 1.396 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	483.
Web	194.4	12.0	6.0	23.3	13998.4	183.
Totals	...	...	...	38.7	130734.4	666.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 140.0 - 12.0
 = 128.000 mm.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 130734.438/38.691
 = 33.790 mm.

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(33.79, 140.0 - 33.79)
 = 106.210 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{666.4/38.691}$
 = 41.500 mm.

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 700.0 - cos(120.0/2)(425.0 + 12.0 + 12.0) - 16.0
 = 459.500 mm.

Intermediate Term [Cc]:
 = $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.20619\text{E}+09/262.0)}$
 = 124.642

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 459.5/41.5
 = 11.072

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 337.9/ (2 • 820.0) • 388.845 • 459.5/2
 = 18.404 Kg-m.

Compressive Allowable, KL/r < Cc (11.0722 < 124.6424) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (11.07)^2 / (2 \cdot 124.64^2)) 262 / (5/3 + 3 \cdot (11.07) / (8 \cdot 124.64) - (11.07^3) / (8 \cdot 124.64^3))$
 = 153.5 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I)/Sba
 = 1.4/153.52 + (18.4 • 106.21/6663611)/174.66
 = 0.026

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	128.0	76.0	15.4	116736.0	636.
Web	388.8	12.0	6.0	46.7	27996.9	146.
Totals	...	...	...	62.0	144732.9	782.



Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 144732.875 / 62.021 \\ &= 23.336 \text{ mm.} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \max(ytot, \text{abs}(\text{Saddle Width} - ytot)) \\ &= \max(23.336, \text{abs}(140.0 - 23.336)) \\ &= 116.664 \text{ mm.} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} &= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk} \\ &= 700.0 - \cos(449.0/2) (16.0 + 0.0 + 0.0) - 0.0 \\ &= 235.000 \text{ mm.} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\ &= \sqrt{ 781.6 / 62.021 } \\ &= 35.498 \text{ mm.} \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL/r \\ &= 1 \cdot 235.0 / 35.498 \\ &= 6.620 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned} &= FI / (2 \cdot \text{Baseplate Length}) \\ &= 337.851 / (2 \cdot 820.0) \\ &= 0.206 \text{ Kg/mm.} \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned} &= \text{Unit Force} \cdot e \cdot L \\ &= 0.206 \cdot 388.845 \cdot 235.0 \\ &= 18.825 \text{ Kg-m.} \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned} &= \text{Bending Moment} / \text{Section Modulus} \\ &= 18.825 / 66992.234 \\ &= 2.756 \text{ N./mm}^2 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ (6.62 < 124.6424) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (6.62)^2 / (2 \cdot 124.64^2)) 262 / \\ &\quad (5/3 + 3 \cdot (6.62) / (8 \cdot 124.64) - (6.62^3) / (8 \cdot 124.64^3)) \\ &= 155.1 \text{ N./mm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} &= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba} \\ &= 1.73 / 155.12 + (18.82 \cdot 116.664 / 781.558) / 174.66 \\ &= 0.027 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

SHELL-2

Design Internal Pressure, Small End	25.057	bars
Design Internal Pressure, Large End	25.057	bars
Temperature for Internal Pressure	85.0	°C
Design External Pressure	1.034	bars
Temperature for External Pressure	85.0	°C

CONE Properties:

Cone Material		SA-516 70	+N+IT
Cone Allowable Stress at Temperature	Ss	138.00	N./mm ²
Cone Allowable Stress At Ambient		138.00	N./mm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	12.0000	mm.
Corrosion Allowance for Cone	Cac	1.5000	mm.
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	410.000	mm.
Diameter of Large End of Cone	Dl	850.000	mm.
Half Apex Angle for Cone		30.00	degrees
Axial Length of Cone	Lc	762.000	mm.

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70	+N+IT
Operating Allowable Stress	Ss	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	12.0000	mm.
Corrosion Allowance	Cas	1.5000	mm.
Axial Length of Attached Element	Ls	100.000	mm.

LARGE-END Properties:

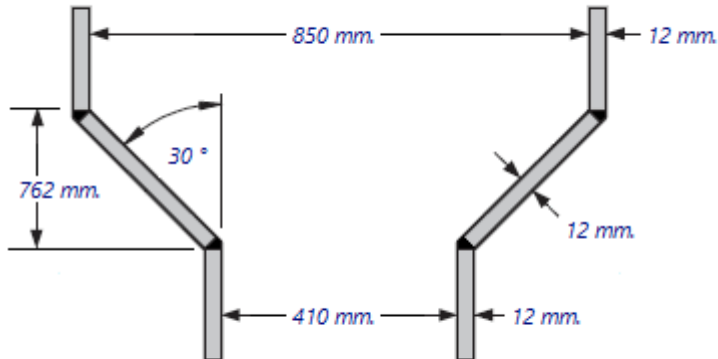
Attached Large End Material Specification		SA-516 70	+N+IT
Operating Allowable Stress	Sl	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	12.0000	mm.
Corrosion Allowance	Cal	1.5000	mm.
Axial Length of Attached Element	Ll	2921.634	mm.

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange No

Force Arl & Ars calculation, cone -> cylinder No



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 85 °C

Elastic Modulus of Cone Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Small Cylinder Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Large Cylinder Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Large End Reinforcement	0.199E+09 KPa. at 85 °C
Elastic Modulus of Small End Reinforcement	0.199E+09 KPa. at 85 °C

Elastic Modulus Data from ASME Section II Part D at 85 °C

Elastic Modulus of Cone Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Small Cylinder Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Large Cylinder Material	0.199E+09 KPa. at 85 °C
Elastic Modulus of Large End Reinforcement	0.199E+09 KPa. at 85 °C
Elastic Modulus of Small End Reinforcement	0.199E+09 KPa. at 85 °C

Axial forces and moments are not computed for Horizontal geometries.
 Compute them manually and enter the loads using the Force/Moment dialog.
 Place the loads at the ends of the cone as needed.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	16.9346 mm.
Maximum Centroid Reinforcement Distance Small End	11.9334 mm.

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Reinforcement Calculations for Cone / Large Cylinder:

Required Area of Reinforcement for Large End Under Internal Pressure

Note: Large end is in compression, 1.0 will be used for E1 and E2.

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.01816

Large End Maximum half apex angle w/o reinforcement [δ]:
 $= 326.6 \cdot \sqrt{P / (Ss \cdot E1)}$ per 1-5 equation 1
 $= 326.6 \cdot \sqrt{25.057 / (138.0 \cdot 1.0)}$
 $= 44.010$ degrees

Large end actual half apex angle [alpha]: 30.000 degrees

Because delta is greater than alpha, reinforcement calculations are not required

for the large end under internal pressure.

Required Area of Reinforcement for Large End Under External Pressure

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.00075

Large End Maximum half apex angle w/o reinforcement [δ]:
 $= 104 \cdot \sqrt{(P / (Ss \cdot E1))}$ per 1-8 equation 1
 $= 104 \cdot \sqrt{(1.034 / (138.0 \cdot 1.0))}$
 $= 2.847$ degrees

Large end actual half apex angle [alpha]: 30.000 degrees

Intermediate Value [k]:
 $= \max(Y / (Srl \cdot Erl), 1)$
 $= \max(0.27468E+11 / (138.0 \cdot 0.19904E+09), 1)$
 $= 1.0000$

where [Y] is:
 $= \text{Large End All. Stress} \cdot \text{Large End Elastic Modulus (Ext. temp.)}$
 $= 138.0 \cdot 0.19904E+09$
 $= 27467829248.0 \text{ (N./mm}^2\text{)}$

Allowable Stress of Large End Material (Ext. Temp)	138.0	N./mm ²
Allowable Stress of Cone Material (Ext. Temp)	138.0	N./mm ²

Required Area of Reinforcement, Large End, External [Arl]:

$= (k \cdot QL \cdot RI \cdot \tan(\text{angle})) / (Ss \cdot E1) \cdot (1 - 1/4 \cdot ((P \cdot RI - QL) / QL) \cdot (\delta / \alpha))$
 $= (1.0 \cdot 2.3043 \cdot 437.0 \cdot 0.577 / (138 \cdot 1.0)) \cdot$
 $(1 - 1/4 \cdot ((1.03 \cdot 437.0 - 2.304) / 2.304) \cdot (2.847 / 30.0))$
 $= 0.403 \text{ cm}^2$

Force per Length, Cone Large End External Pressure, compressive side [QL-]:

$= P_{ext}(RI/2) + f1$
 $= 1.034(437.0/2) + 0.0$
 $= 2.304 \text{ Kg/mm.}$

Axial Load per Unit Circumference excluding Pressure [f1]:

$= F_{axial} / (\pi(DI + TI)) + \text{Moment} / (\pi(RI - TI/2)^2)$
 $= +.0 / (\pi(874.0 - 12.0)) + .0 / (\pi(437.0 - 12.0/2)^2)$
 $= 0.0 \text{ Kg/mm.}$

Force per Length, Cone Large End External Pressure, tensile side [QL+]:

$= P_{ext}(RI/2) + f1$
 $= 1.034(437.0/2) + 0.0$
 $= 2.304 \text{ Kg/mm.}$

Axial Load per Unit Circumference excluding Pressure [f1]:

$= F_{axial} / (\pi(DI + TI)) - \text{Moment} / (\pi(RI - TI/2)^2)$
 $= +.0 / (\pi(874.0 - 12.0)) - 0.0 / (\pi(437.0 - 12.0/2)^2)$
 $= 0.0 \text{ Kg/mm.}$

Available Area of Reinforcement, Large End, External [Ael]:

$= 0.55[\sqrt{(DI \cdot ts)(ts - t)} + \sqrt{(DI \cdot tc / \cos(\alpha))(tc - tr)}]$ per 1-8 eqn. 3
 $= 0.55[\sqrt{(874.0 \cdot 10.5)(10.5 - 4.629)} +$
 $\sqrt{(874.0 \cdot 10.5/0.866)(10.5 - 2.711)}]$
 $= 7.5027 \text{ cm}^2$

Summary of Reinforcement Area, Large End, External Pressure:

Area of reinforcement required per App. 1-8(1)	0.4034	cm ²
Area of reinforcement in shell per App. 1-8(2)	7.5027	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²

Intermediate Results, Large End, External Pressure:

Area Available in Cone, Shell, and Reinforcement [AT]:

$= LL \cdot ts/2 + Lc \cdot tc/2 + As$
 $= 2921.634 \cdot 10.5/2 + 793.123 \cdot 10.5/2 + 0.0$
 $= 195.025 \text{ cm}^2$



Determine the Meridian Length [M]:

$$= -RL \cdot \tan(\alpha)/2 + LL/2 + (RL^2 - RS^2)/(3 \cdot RL \cdot \tan(\alpha))$$

$$= -437.0 \cdot \tan(30.0)/2 + 2921.634/2 + (437.0^2 - 217.0^2)/(3 \cdot 437.0 \cdot \tan(30.0))$$

$$= 1524.755 \text{ mm.}$$

Force per Unit Length on the Cone/Shell Junction [FL]:

$$= P \cdot M + \tan(\alpha)(\text{Force}/(\pi \cdot D) + \text{Moment}/(\pi \cdot \text{Radius}^2))$$

$$= 1.034 \cdot 1524.8 + \tan(30.0)(0.0/(\pi \cdot 874.0) + 0.0/(\pi \cdot 437.0^2))$$

$$= 16.080 \text{ Kg/mm.}$$

Actual Buckling Stress associated with force FL, [B]:

$$= 0.75(FL \cdot DL / ATL)$$

$$= 0.75(16.08 \cdot 874.0 / 195.025)$$

$$= 5.300 \text{ N./mm}^2$$

Material Strain associated with this stress [A]: 0.000053

Required Moment of Inertia, Large End, External Pressure [I's]:

$$= A \cdot DI^2 \cdot Atl / 10.9$$

$$= 0.533E-04 \cdot 874.0 \cdot 874.0 \cdot 195.02/10.9$$

$$= 72787.09 \text{ mm.}^4$$

Available Moment of Inertia, Large End, External Pressure:

	Area	Centroid	Ar*Ce	Dist	I	Ar*Di ²
Shl	5.532	0.0000	0.000	-8.1509	0.508	367.546
Con	6.388	-15.2098	-97.162	7.0589	5.709	318.305
Sec	0.000	5.2500	0.000	-13.4009	0.000	0.000
TOT	11.920		-97.162		6.217	685.851

Centroid of Section -8.1509 Moment of Inertia 13.08

Summary of Large End Inertia Calculations

Available Moment of Inertia (Large End)	13.075	cm ⁴
Required Moment of Inertia (Large End)	7.279	cm ⁴

Reinforcement Calculations for Cone / Small Cylinder:

Required Area of Reinforcement for Small End under Internal Pressure

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.01816

Small End Maximum half apex angle w/o reinforcement [δ]:

$$= 89 \cdot \sqrt{(P / (Ss \cdot E1))} \text{ per 1-5 equation 6}$$

$$= 89 \cdot \sqrt{(25.057 / (138.0 \cdot 1.0))}$$

$$= 11.993 \text{ degrees}$$

Small end actual half apex angle [alpha]: 30.000 degrees

Intermediate Value [k]:

$$= \max(Y / (Sr \cdot Ers), 1)$$

$$= \max(0.27468E+11 / (138.0 \cdot 0.19904E+09), 1)$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Int. temp.)}$$

$$= 138.0 \cdot 0.19904E+09$$

$$= 27467829248.0 \text{ (N./mm}^2\text{)}^2$$

Decay Length, Cone Small End:

$$= 1.4 \cdot \sqrt{(Rs / (ts - ca))}$$

$$= 1.4 \cdot \sqrt{(206.5 / (12.0 - 1.5))}$$

$$= 65.190 \text{ mm.}$$

Required Area of Reinforcement, Small End, Internal [Ars]:

$$= k \cdot QS \cdot Rs / (Ss \cdot E1) \cdot (1 - \delta/\alpha) \cdot \tan(\alpha) \text{ per 1-5 equation 7}$$

$$= 1.0 \cdot 26.3823 \cdot 206.5 / (138 \cdot 1.0) \cdot (1 - 11.99/30.0) \cdot 0.5774$$

$$= 1.342 \text{ cm}^2$$

Force per Length, Cone Small End, tensile side [QS+]:

$$= P(Rs/2) + f2$$

$$= 25.057(206.5/2) + 0.0$$

$$= 26.382 \text{ Kgf/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) + \text{Moment}/(\pi(Rs + Ts/2)^2)$$

$$= -0.0/(\pi(413.0+12.0)) + 0.0/(\pi(206.5+12.0/2)^2)$$

$$= 0.0 \text{ Kgf/mm.}$$

Force per Length, Cone Small End, compressive side [QS-]:

$$= P(Rs/2) + f2$$

$$= 25.057(206.5/2) + -0.0$$

$$= 26.382 \text{ Kgf/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) - \text{Moment}/(\pi(Rs + Ts/2)^2)$$

$$= -0.0/(\pi(413.0+12.0)) - 0.0/(\pi(206.5+12.0/2)^2)$$

$$= -0.0 \text{ Kgf/mm.}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.78[\sqrt{(Rs \cdot ts)} \cdot (ts - t) + \sqrt{(Rs \cdot tc/\cos(\alpha))} \cdot (tc - tr)] \text{ per 1-5 eqn. 8}$$

$$= 0.78[\sqrt{(206.5 \cdot 10.5)} \cdot (10.5 - 3.791) + \sqrt{(206.5 \cdot 10.5/0.866)} \cdot (10.5 - 4.377)]$$

$$= 4.8263 \text{ cm}^2$$

Summary of Reinforcement Area, Small End, Internal Pressure:

Area of reinforcement required per App. 1-5(7)	1.3417	cm ²
Area of reinforcement in shell per App. 1-5(8)	4.8263	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²

Required Area of Reinforcement for Small End Under External Pressure

Note: Small end is in compression, 1.0 will be used for E1 and E2.

Allowable Stress of Small End Material (Ext. Temp)	138.0	N./mm ²
Allowable Stress of Cone Material (Ext. Temp)	138.0	N./mm ²

Intermediate Value [k]:

$$= \max(Y / (Srs \cdot Ers), 1)$$

$$= \max(0.27468E+11 / (138.0 \cdot 0.19904E+09), 1)$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Ext. temp.)}$$

$$= 138.0 \cdot 0.19904E+09$$

$$= 27467829248.0 \text{ (N./mm}^2\text{)}$$

Area of Reinforcement Required in Small End Shell [Ars]:

$$= k \cdot QS \cdot Rs \cdot \tan(\alpha) / (Ss \cdot E1)$$

$$= (1.0 \cdot 1.1443 \cdot 217.0 \cdot 0.5774) / (138 \cdot 1.0)$$

$$= 0.102 \text{ cm}^2$$

Force per Length, Cone Small End External Pressure, compressive side [QS-]:

$$= \text{Pext}(Rs/2) + f2$$

$$= 1.034(217.0/2) + 0.0$$

$$= 1.144 \text{ Kgf/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) + \text{Moment}/(\pi(Rs - Ts/2)^2)$$

$$= +0.0/(\pi(434.0-12.0)) + 0.0/(\pi(217.0-12.0/2)^2)$$

$$= 0.0 \text{ Kgf/mm.}$$

Force per Length, Cone Small End External Pressure, tensile side [QS+]:

$$= \text{Pext}(Rs/2) + f2$$

$$= 1.034(217.0/2) + 0.0$$

$$= 1.144 \text{ Kgf/mm.}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= \text{Faxial}/(\pi(Ds + Ts)) - \text{Moment}/(\pi(Rs - Ts/2)^2)$$

$$= +0.0/(\pi(434.0-12.0)) - 0.0/(\pi(217.0-12.0/2)^2)$$

$$= 0.0 \text{ Kgf/mm.}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.55 \left[\sqrt{(Ds \cdot ts)(ts-t)} + \sqrt{(Ds \cdot tc / \cos(\alpha))(tc-tr)} \right]$$

$$= 0.55 \left[\sqrt{(434.0 \cdot 10.5)(10.5 - 46.15)} + \sqrt{(434.0 \cdot 10.5 / 0.866)(10.5 - 1.809)} \right]$$

$$= 3.4675 \text{ cm}^2$$

Summary of Reinforcement Area, Small End, External Pressure:

Area of reinforcement required per App. 1-8(6)	0.1019	cm ²
Area of reinforcement in shell per App. 1-8(7)	3.4675	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²

Intermediate Results, Small End, External Pressure:

Area Available in Cone, Shell, and Reinforcement	46.89	cm ²
Force per Unit Length on Shell / Cone Junction	3.21	Kgf/mm.
Actual Buckling Stress associated with this Force	2.18	N./mm ²
Material Strain associated with this stress	0.000022	

Required Moment of Inertia, Small End, External Pressure [I's]:

$$= A \cdot Ds^2 \cdot Ats / 10.9$$

$$= 0.219\text{E-}04 \cdot 434.0 \cdot 434.0 \cdot 46.89 / 10.9$$

$$= 1777.17 \text{ mm}^4$$

Available Moment of Inertia, Small End, External Pressure:

	Area	Centroid	Ar*Ce	Dist	I	Ar*Di ²
Shl	3.898	0.0000	0.000	5.7437	0.358	128.611
Con	4.502	10.7179	48.247	-4.9742	2.275	111.381
Sec	0.000	5.2500	0.000	0.4937	0.000	0.000
TOT	8.400		48.247		2.633	239.992

Centroid of Section	5.7437	Moment of Inertia	5.03
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Summary of Small End Inertia Calculations

Available Moment of Inertia (Small End)	0.5033E+01	cm ⁴
Required Moment of Inertia (Small End)	0.1777E+00	cm ⁴



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	98.99	484.00	Small Cyl. Long.
Compres. Stress	-48.45	-484.00	Small Cyl. Long.
Membrane Stress	91.11	-207.00	Small End Tang.
Tensile Stress	102.89	484.00	Cone Longitudinal
Compres. Stress	-44.54	-484.00	Cone Longitudinal
Compres Stress	98.93	-207.00	Cone Tangential
Tensile Stress	266.15	484.00	Large Cyl. Long.
Compres. Stress	-163.11	-484.00	Large Cyl. Long.
Membrane Stress	-15.10	-207.00	Large End Tang.
Tensile Stress	274.12	484.00	Cone Longitudinal
Compres. Stress	-155.14	-484.00	Cone Longitudinal
Compres Stress	0.84	-207.00	Cone Tangential

An asterisk (*) denotes that this stress was not applicable for this combination of loads.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure bars	Reason for Failure at this Pressure
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MAWP	29.052	Thickness due to internal pressure, Cone Large End
MAPnc	33.261	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Input, Nozzle Desc: TS IN NOZ. A1

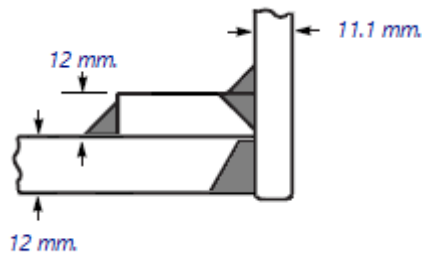
From: 20

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	410.00	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	188.9000	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		50.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS IN NOZ. A1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.624 in.
 Actual Thickness Used in Calculation 0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 206.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 3.7823$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.0 \cdot 34.82 / (118 \cdot 1.0 - 0.6 \cdot 25.0)$
 $= 0.7474$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	139.2992 mm.
Parallel to Vessel Wall, opening length	d	69.6496 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		26.2500 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: TS IN NOZ. A1.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.2474 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 5.2823 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.2823 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 6.3000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(5.2823, 3.0)]$
 $= 5.2823$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.2474, 5.2823)$
 $= 5.2823$ mm.

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735$ mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	31.0,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	264.0 N./mm ²	Passed



Occasional : 4.0, Allowable : 156.9 N./mm² Passed
Shear : 19.1, Allowable : 82.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
Calculated Minimum Design Metal Temperature -104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
Calculated Minimum Design Metal Temperature -104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
Calculated Minimum Design Metal Temperature -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2) -104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = 25.00/102.00 = 0.245

Weld Size Calculations, Description: TS IN NOZ. A1

Intermediate Calc. for nozzle/shell Welds Tmin 9.6252 mm.
Intermediate Calc. for pad/shell Welds TminPad 10.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	5.2500 = 0.5*TminPad	7.0700 = 0.7 * Wp mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.



Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 68 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 4.8771 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.8770 mm.

Input Echo, WRC297 Item 1, Description: TS IN NOZ. A1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	410.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	66.650	mm.
Nozzle Thickness	11.1252	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	188.900	mm.
Thickness of Reinforcing Pad	12.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	25.00	bars
Radial Load	P	-458.90	Kgf
Circumferential Shear	Vc	344.18	Kgf
Longitudinal Shear	Vl	458.90	Kgf
Circumferential Moment	Mc	89.45	Kg-m.
Longitudinal Moment	Ml	89.45	Kg-m.
Torsional Moment	Mt	103.21	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (7.991) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(217.75(12.0 - 1.5))}$$

$$= 246.693 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	435.500	mm.
Nozzle Outside Diameter	(d) =	88.900	mm.
Vessel Thickness used	(T) =	22.500	mm.
Nozzle Thickness used	(t) =	9.625	mm.



T / t = 2.338
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.898$$

Nr/P = 0.153
 Mr/P = 0.128
 M0/P = 0.094
 N0/P = 0.164
 MrD/Mc = 0.204
 NrDL/Mc = 0.189
 M0d/Mc = 0.170
 N0DL/Mc = -0.055
 MrD/Ml = 0.164
 NrDL/Ml = 0.156
 M0D/Ml = 0.140
 N0DL/Ml = 0.145

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	21	24	21	24
Outplane Stress Summary	8	34	47	7

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-19	19	19	-19
Inplane Pressure Stress	10	10	10	10
Inplane Stress Summary	-3	21	41	-10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	15	35	48	18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-19	19	19	-19
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	10	10	10	10
Outplane Stress Summary	-1	28	36	-13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-23	23	23	-23
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	21	24	21	24
Inplane Stress Summary	2	38	57	-1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	5	39	58	15

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	12	9	12
Hoop Stress Summary	8	11	13	16

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	15	-15	15	-15
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-55	55	55	-55



Axial Pressure Stress	4	4	4	4
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Axial Stress Summary	-53	26	97	-43
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	10	10	5	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	65	31	97	61

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	12	9	12
Hoop Stress Summary	12	15	9	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	15	-15	15	-15
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-69	69	69	-69
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-68	40	112	-57

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	8	8	8	8
Shear Stress Summary	5	5	11	11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	81	41	113	74

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:



$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(211.75(12.0 - 1.5))}$$

$$= 244.504 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 423.500 mm.
 Nozzle Outside Diameter (d) = 188.900 mm.
 Vessel Thickness used (T) = 10.500 mm.
 Nozzle Thickness used (t) = 59.625 mm.
 T / t = 0.176
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.833$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.061
 Mr/P = 0.112
 M0/P = 0.033
 N0/P = 0.059
 MrD/Mc = 0.271
 NrDL/Mc = 0.096
 M0d/Mc = 0.081
 N0DL/Mc = 0.080
 MrD/M1 = 0.086
 NrDL/M1 = 0.073
 M0D/M1 = 0.025
 N0DL/M1 = 0.075

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-6	6	6	-6
Normal Pressure Stress	47	50	47	50
Outplane Stress Summary	49	47	67	41

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-21	21	21	-21
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	28	17	78	-19

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	49	48	78	61

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-20	20	20	-20
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23
Outplane Stress Summary	10	35	58	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	27	-27	27	-27
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-68	68	68	-68
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	47	50	47	50
Inplane Stress Summary	5	90	150	-38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	10	90	150	40

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	24	21	24	21	24	21	24
Circ. Pl (SUS)	-1	-1	3	3	-2	-2	4	4
Circ. Q (SUS)	-11	11	21	-21	-17	17	29	-29
Long. Pm (SUS)	10	10	10	10	10	10	10	10

Long. P1 (SUS)	-2	-2	4	4	2	2	0	0
Long. Q (SUS)	-13	13	25	-25	-14	14	24	-24
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	21.0	24.0	21.0	24.0	21.0	24.0	21.0	24.0
Pm+P1 (SUS)	20.1	23.1	24.1	27.1	19.1	22.1	25.1	28.1
Pm+P1+Q (Total)	16.1	35.1	45.6	17.5	5.7	39.3	54.8	15.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	138.00	Passed
Pm+P1 (SUS)	28.06	207.00	Passed
Pm+P1+Q (TOTAL)	54.77	414.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	47	50	47	50	47	50	47	50
Circ. P1 (SUS)	-1	-1	5	5	-2	-2	6	6
Circ. Q (SUS)	2	-2	14	-14	-41	41	95	-95
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. P1 (SUS)	-1	-1	5	5	-1	-1	5	5
Long. Q (SUS)	6	-6	48	-48	-12	12	28	-28
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	47.0	50.0	47.0	50.0	47.0	50.0	47.0	50.0
Pm+P1 (SUS)	46.0	49.0	52.0	55.0	45.0	48.0	53.0	56.0
Pm+P1+Q (Total)	48.2	47.1	76.0	61.0	10.0	89.0	148.0	39.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	50.00	138.00	Passed
Pm+P1 (SUS)	56.04	207.00	Passed
Pm+P1+Q (TOTAL)	148.04	414.00	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	9	12	9	12	9	12	9	12
Circ. Pl (SUS)	-1	-1	3	3	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-19	-19	21	21	-19	-19	21	21
Long. Q (SUS)	-40	40	70	-70	-54	54	84	-84
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	9.0	12.0	9.0	12.0	9.0	12.0	9.0	12.0
Pm+Pl (SUS)	23.3	26.3	25.3	25.4	26.7	29.6	25.5	25.7
Pm+Pl+Q (Total)	66.1	30.2	95.4	61.2	80.6	40.0	110.2	74.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	118.00	Passed
Pm+Pl (SUS)	29.61	177.00	Passed
Pm+Pl+Q (TOTAL)	110.20	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: TS OUT NOZ. B1

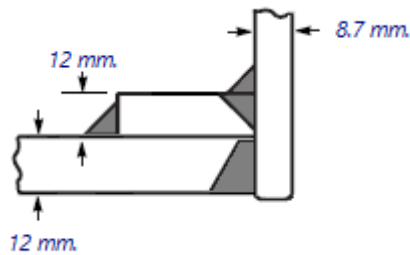
From: 20

Pressure for Reinforcement Calculations	P	25.027	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	410.00	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		50.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS OUT NOZ. B1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.03 \cdot 206.5 / (138 \cdot 1.0 - 0.6 \cdot 25.03)$
 $= 3.7864$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.03 \cdot 22.92 / (118 \cdot 1.0 - 0.6 \cdot 25.03)$
 $= 0.4925$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	91.6996 mm.
Parallel to Vessel Wall, opening length	d	45.8498 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		26.2500 mm.

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: TS OUT NOZ. B1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9925 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 5.2864 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 5.2864 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 4.9200 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(5.2864, 3.0)]$
 $= 4.9200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9925, 4.92)$
 $= 4.9200$ mm.

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	39.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	255.7 N./mm ²	Passed



Occasional : 3.4, Allowable : 156.9 N./mm² Passed
 Shear : 26.0, Allowable : 82.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
 Calculated Minimum Design Metal Temperature -104 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C
 Calculated Minimum Design Metal Temperature -104 °C
 Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: TS OUT NOZ. B1

Intermediate Calc. for nozzle/shell Welds Tmin 7.2376 mm.
 Intermediate Calc. for pad/shell Welds TminPad 10.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.0663 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.
Pad Weld	5.2500 = 0.5*TminPad	7.0700 = 0.7 * Wp mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 2.2311 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 214.2311 mm.

Input Echo, WRC297 Item 1, Description: TS OUT NOZ. B1:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	410.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	S _{mc}	138.00 N./mm ²
Vessel Hot S.I. Allowable	S _{mh}	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.



Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	42.850	mm.
Nozzle Thickness	8.7376	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	160.325	mm.
Thickness of Reinforcing Pad	12.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	25.03	bars
Radial Load	P	-305.94	Kgf
Circumferential Shear	Vc	229.45	Kgf
Longitudinal Shear	Vl	305.94	Kgf
Circumferential Moment	Mc	39.76	Kg-m.
Longitudinal Moment	Ml	39.76	Kg-m.
Torsional Moment	Mt	45.87	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (6.904) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(217.75(12.0 - 1.5))}$
 $= 218.118 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	435.500	mm.
Nozzle Outside Diameter	(d) =	60.325	mm.
Vessel Thickness used	(T) =	22.500	mm.
Nozzle Thickness used	(t) =	7.238	mm.
T / t	=	3.109	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.609$$

Nr/P	=	0.184
Mr/P	=	0.149
M0/P	=	0.162
N0/P	=	0.193
MrD/Mc	=	0.182
NrDL/Mc	=	0.183
M0d/Mc	=	0.217
N0DL/Mc	=	-0.088
MrD/Ml	=	0.167
NrDL/Ml	=	0.173
M0D/Ml	=	0.201
N0DL/Ml	=	0.055

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	21	24	21	24

Outplane Stress Summary	12	34	44	4
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-12	12	12	-12
Inplane Pressure Stress	10	10	10	10

Inplane Stress Summary	2	17	32	-3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	14	35	45	10
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-16	16	16	-16
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	10	10	10	10

Outplane Stress Summary	2	23	33	-11
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-13	13	13	-13
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0



Inplane Pressure Stress	21	24	21	24
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Inplane Stress Summary	11	31	44	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	12	32	46	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	11	8	11
Hoop Stress Summary	8	11	10	13

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-27	-27	27	27
Axial Bending (ML)	-59	59	59	-59
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-60	19	112	-45

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	14	14	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	74	30	113	60

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses in the hoop direction)	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	8	11	8	11
Hoop Stress Summary	10	13	8	11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-27	-27	27	27
Axial Bending (Mc)	-67	67	67	-67
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-68	27	120	-53

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	10	10	10	10
Shear Stress Summary	6	6	15	15

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	80	29	122	71

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(211.75(12.0 - 1.5))}$
 $= 215.929 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 423.500 mm.
 Nozzle Outside Diameter (d) = 160.325 mm.
 Vessel Thickness used (T) = 10.500 mm.
 Nozzle Thickness used (t) = 57.238 mm.
 $T / t = 0.183$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.404$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.066
 Mr/P = 0.127
 M0/P = 0.038
 N0/P = 0.067
 MrD/Mc = 0.281



NrDL/Mc = 0.094
 MØd/Mc = 0.085
 NØDL/Mc = 0.080
 MrD/Ml = 0.098
 NrDL/Ml = 0.082
 MØD/Ml = 0.029
 NØDL/Ml = 0.088

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-3	3	3	-3
Normal Pressure Stress	47	50	47	50
Outplane Stress Summary	50	48	61	44

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-12	12	12	-12
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	31	16	61	-6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	50	48	61	50

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-11	11	11	-11
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23
Outplane Stress Summary	18	29	45	10

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-37	37	37	-37
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	47	50	47	50
Inplane Stress Summary	31	66	109	-3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	31	66	109	14

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	24	21	24	21	24	21	24
Circ. Pl (SUS)	1	1	1	1	-1	-1	3	3
Circ. Q (SUS)	-10	10	20	-20	-8	8	18	-18
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-1	-1	3	3	2	2	0	0
Long. Q (SUS)	-7	7	17	-17	-11	11	21	-21
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	21.0	24.0	21.0	24.0	21.0	24.0	21.0	24.0
Pm+Pl (SUS)	22.1	25.1	22.1	25.1	20.1	23.1	24.1	27.1
Pm+Pl+Q (Total)	13.4	35.8	42.3	9.8	12.4	31.5	43.3	21.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	138.00	Passed
Pm+Pl (SUS)	27.06	207.00	Passed
Pm+Pl+Q (TOTAL)	43.30	414.00	Passed



Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	47	50	47	50	47	50	47	50
Circ. Pl (SUS)	0	0	2	2	-1	-1	3	3
Circ. Q (SUS)	3	-3	9	-9	-17	17	57	-57
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	0	0	2	2	0	0	2	2
Long. Q (SUS)	8	-8	32	-32	-5	5	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	47.0	50.0	47.0	50.0	47.0	50.0	47.0	50.0
Pm+Pl (SUS)	47.0	50.0	49.0	52.0	46.0	49.0	50.0	53.0
Pm+Pl+Q (Total)	50.1	47.0	58.6	50.0	29.0	66.0	107.1	12.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	50.00	138.00	Passed
Pm+Pl (SUS)	53.04	207.00	Passed
Pm+Pl+Q (TOTAL)	107.06	414.00	Passed

Because only sustained loads were specified, the $Pm+Pl+Q$ allowable was $3 * Smh$.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	8	11	8	11	8	11	8	11
Circ. Pl (SUS)	1	1	1	1	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-25	-25	29	29	-25	-25	29	29
Long. Q (SUS)	-40	40	78	-78	-48	48	86	-86
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-4	-4	4	4
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	8.0	11.0	8.0	11.0	8.0	11.0	8.0	11.0
Pm+Pl (SUS)	30.6	33.5	33.4	33.4	32.0	34.9	33.6	33.7
Pm+Pl+Q (Total)	74.7	29.0	111.5	58.7	79.9	29.2	120.7	69.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	11.00	118.00	Passed
Pm+Pl (SUS)	34.93	177.00	Passed
Pm+Pl+Q (TOTAL)	120.74	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: SS IN NOZ A2

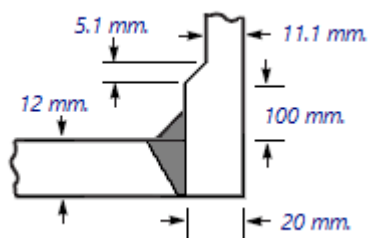
From: 70

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	850.00	mm.
Design Length of Section	L	2921.6335	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3910.18	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	11.1300	mm.
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	100.0000	mm.
Height of Beveled Transition	L'	5.1211	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: SS IN NOZ A2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
Actual Thickness Used in Calculation 0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 426.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 7.8118 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 39.6 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 0.7253 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4234 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
= 2.5 • Corroded Hub Thickness
= 2.5 • 18.5, less than the hub height, use (e-2)
= 46.2500 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.4000	mm.
Parallel to Vessel Wall, opening length	d	79.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: SS IN NOZ A2.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.2253	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.3118	mm.
Wall Thickness	tb1	= max(trb1, tr16b)	= 9.3118 mm.
Wall Thickness, shell/head, external pressure	trb2	= 1.8198	mm.
Wall Thickness	tb2	= max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 6.5200	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.52, max(9.3118, 3.0)]
= 6.5200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.2253, 6.52)
= 6.5200 mm.

Available Nozzle Neck Thickness = 11.1300 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	25.7,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	319.3 N./mm ²	Passed
Occasional	:	4.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	15.4,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of



the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Calculated Minimum Design Metal Temperature -104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C

Flange MDMT with Temp reduction per UCS-66(i)(2) -104 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

Design Pressure/Ambient Rating = 25.00/102.00 = 0.245

Weld Size Calculations, Description: SS IN NOZ A2

Intermediate Calc. for nozzle/shell Welds Tmin 12.5000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 33 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.9900 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 215.9900 mm.

Input Echo, WRC297 Item 1, Description: SS IN NOZ A2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	850.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	76.200	mm.
Nozzle Thickness	20.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	



Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	25.00	bars
Radial Load	P	-458.90	Kgf
Circumferential Shear	Vc	344.18	Kgf
Longitudinal Shear	Vl	458.90	Kgf
Circumferential Moment	Mc	89.45	Kg-m.
Longitudinal Moment	Ml	89.45	Kg-m.
Torsional Moment	Mt	103.21	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (5.810) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 116.2 + 2 \cdot 1.65 \cdot \sqrt{(431.75(12.0 - 1.5))}$$

$$= 338.390 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	863.500 mm.
Nozzle Outside Diameter	(d) =	116.200 mm.
Vessel Thickness used	(T) =	10.500 mm.
Nozzle Thickness used	(t) =	18.500 mm.
T / t	=	0.568
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.220$$

Nr/P	=	0.096
Mr/P	=	0.193
M0/P	=	0.057
N0/P	=	0.122
MrD/Mc	=	0.300
NrDL/Mc	=	0.101
M0d/Mc	=	0.096
N0DL/Mc	=	0.048
MrD/Ml	=	0.171
NrDL/Ml	=	0.086
M0D/Ml	=	0.056
N0DL/Ml	=	0.129

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-8	-8	8	8
Outplane Bending (Ml)	-23	23	23	-23
Normal Pressure Stress	100	102	100	102
Outplane Stress Summary	87	108	151	79



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-70	70	70	-70
Inplane Pressure Stress	50	50	50	50
Inplane Stress Summary	25	71	177	-57

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	87	109	177	137

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-39	39	39	-39
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	50	50	50	50
Outplane Stress Summary	26	77	111	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-123	123	123	-123
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	100	102	100	102
Inplane Stress Summary	21	175	281	-56

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0



Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	27	175	281	63

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8
Hoop Stress Summary	1	4	19	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-16	16	16	-16
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-9	1	39	-17

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	12	7	39	39

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8

Hoop Stress Summary 6 9 13 16

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	11	-11	11	-11
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-32	32	32	-32
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary -25 17 55 -33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	2	2	2	2

Shear Stress Summary 1 1 3 3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	32	17	55	50

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	100	102	100	102	100	102	100	102
Circ. Pl (SUS)	-4	-4	12	12	-3	-3	9	9
Circ. Q (SUS)	-9	9	37	-37	-76	76	170	-170
Long. Pm (SUS)	50	50	50	50	50	50	50	50
Long. Pl (SUS)	-2	-2	8	8	1	1	7	7
Long. Q (SUS)	-23	23	117	-117	-25	25	53	-53
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-2	-2	2	2
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	100.0	102.0	100.0	102.0	100.0	102.0	100.0	102.0
Pm+Pl (SUS)	96.0	98.0	112.0	114.0	97.1	99.1	109.1	111.1
Pm+Pl+Q (Total)	87.4	107.7	175.3	136.1	26.7	175.0	279.2	64.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	102.00	138.00	Passed
Pm+Pl (SUS)	114.02	207.00	Passed
Pm+Pl+Q (TOTAL)	279.21	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	8	5	8	5	8	5	8
Circ. Pl (SUS)	-4	-4	12	12	1	1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-5	5	27	-27	-21	21	43	-43
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	5.0	8.0	5.0	8.0	5.0	8.0	5.0	8.0
Pm+Pl (SUS)	6.0	9.0	17.0	20.0	11.2	14.1	12.3	15.2
Pm+Pl+Q (Total)	11.7	5.7	36.2	38.2	32.1	16.1	52.2	49.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	20.00	207.00	Passed
Pm+Pl+Q (TOTAL)	52.22	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: SS OUT NOZ B2

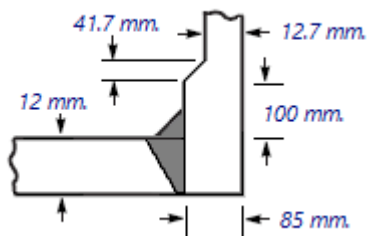
From: 70

Pressure for Reinforcement Calculations	P	25.000	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	850.00	mm.
Design Length of Section	L	2921.6335	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2810.18	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm.
Flange Material		SA-350 LF2	+IT
Hub Height of Integral Nozzle	h	100.0000	mm.
Height of Beveled Transition	L'	41.7424	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		85.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: SS OUT NOZ B2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 8.000 in.
Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 426.5 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 7.8118 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.0 \cdot 103.1 / (138 \cdot 1.0 - 0.6 \cdot 25.0)$
= 1.8884 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.6915 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= 72.3/0.5773
= 125.2273 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

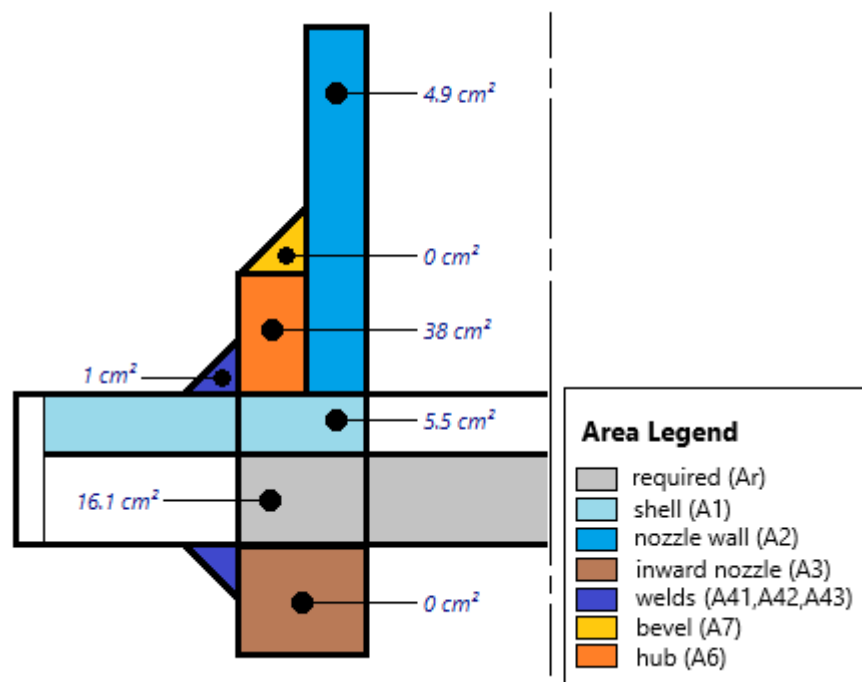
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	412.4000	mm.
Parallel to Vessel Wall, opening length	d	206.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.2500	mm.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n/S_v)$
= $\min(1, 138.0/138.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n/S_v)$
= $\min(1, 138.0/138.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1., 1.0)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	16.108	4.773	NA
Area in Shell	A1	5.543	12.105	NA
Area in Nozzle Wall	A2	4.888	5.517	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	37.957	37.957	NA
TOTAL AREA AVAILABLE	Atot	49.388	56.578	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (206.2 \cdot 7.8118 \cdot 1.0 + 2 \cdot 83.5 \cdot 7.8118 \cdot 1.0 \cdot (1 - 1.))$$

$$= 16.108 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 206.2(1.0 \cdot 10.5 - 1.0 \cdot 7.812) - 2 \cdot 83.5$$

$$(1.0 \cdot 10.5 - 1.0 \cdot 7.8118) \cdot (1 - 1.)$$

$$= 5.543 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 26.25)(11.2 - 1.89) \cdot 1.$$

$$= 4.888 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$

$$= 1.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2 \\ &= (2 \cdot \min(26.2, 200.0, 100.0)) \cdot (85.0 - 12.7) \cdot 1.) \\ &= 37.957 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3884 mm.
Wall Thickness per UG16(b),	tr16b = 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 9.3118 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 9.3118 mm.
Wall Thickness, shell/head, external pressure	trb2 = 1.8198 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.0000 mm.
Wall Thickness per table UG-45	tb3 = 9.6026 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.603, \max(9.3118, 3.0)] \\ &= 9.3118 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(3.3884, 9.3118) \\ &= 9.3118 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 12.7000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.6,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 313.4 N./mm ²	Passed
Occasional	: 10.9,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 13.8,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-46 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-79 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 25.00/51.00 = 0.490$$

Weld Size Calculations, Description: SS OUT NOZ B2

Intermediate Calc. for nozzle/shell welds	Tmin	12.5000 mm.
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Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (16.1082 - 5.543 + 2 \cdot 83.5 \cdot 1. \cdot (1.0 \cdot 10.5 - 7.8118)) \cdot 138)$
 $= 21184.33 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (42.8453 + 1. - 0.0 \cdot 1.) \cdot 138$
 $= 61698.32 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (42.8453 + 0.0 + 1. + (17.5347)) \cdot 138$
 $= 86372.81 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (42.8453 + 0.0 + 1. + 0.0 + (17.5347)) \cdot 138$
 $= 86372.81 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 373.2 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 40420. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 144.85) \cdot (85.0 - 1.5) \cdot 0.7 \cdot 138$
 $= 374279. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 373.2 \cdot (12.0 - 1.5) \cdot 0.74 \cdot 138$
 $= 64096. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (40420 + 374279) = 414699 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (40420 + 0 + 64096 + 0) = 104517 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (40420 + 64096 + 0) = 104517 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 414699 Kgf, must exceed W = 21184 Kgf	or W1 = 61698 Kgf
Path 2-2 = 104516 Kgf, must exceed W = 21184 Kgf	or W2 = 86372 Kgf
Path 3-3 = 104516 Kgf, must exceed W = 21184 Kgf	or W3 = 86372 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 33 bars



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 43.1552 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 255.1552 mm.

Input Echo, WRC297 Item 1, Description: SS OUT NOZ B2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	850.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	203.200	mm.
Nozzle Thickness	85.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	25.00	bars
Radial Load	P	-1223.74	Kgf
Circumferential Shear	Vc	917.81	Kgf
Longitudinal Shear	Vl	1223.74	Kgf
Circumferential Moment	Mc	636.10	Kg-m.
Longitudinal Moment	Ml	636.10	Kg-m.
Torsional Moment	Mt	733.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (4.391) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 373.2 + 2 \cdot 1.65 \cdot \sqrt{(431.75(12.0 - 1.5))}$$

$$= 595.390 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	863.500	mm.
Nozzle Outside Diameter	(d) =	373.200	mm.
Vessel Thickness used	(T) =	10.500	mm.
Nozzle Thickness used	(t) =	83.500	mm.
T / t	=	0.126	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.919$$

Nr/P = 0.052
 Mr/P = 0.085
 M0/P = 0.025
 N0/P = 0.040
 MrD/Mc = 0.254
 NrDL/Mc = 0.100
 M0d/Mc = 0.076
 N0DL/Mc = 0.067
 MrD/M1 = 0.064
 NrDL/M1 = 0.058
 M0D/M1 = 0.019
 N0DL/M1 = 0.044

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	100	102	100	102
Outplane Stress Summary	96	101	144	80

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	55	-55	55	-55
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	50	50	50	50
Inplane Stress Summary	44	49	178	-48

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	97	102	178	128

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-10	-10	10	10



Outplane Bending (Mc)	-69	69	69	-69
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	50	50	50	50

Outplane Stress Summary	-8	97	149	-20
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	55	-55	55	-55
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-231	231	231	-231
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	100	102	100	102

Inplane Stress Summary	-85	269	407	-162
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	85	269	407	163
------------------------	----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5

Hoop Stress Summary	0	3	13	16
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	0	3	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	1	3	13	16

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	5	2	5
Hoop Stress Summary	-3	0	17	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-2	2	2	-2
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1
Axial Stress Summary	-2	2	6	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	3	2	17	20

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	100	102	100	102	100	102	100	102
Circ. Pl (SUS)	-2	-2	10	10	-10	-10	20	20
Circ. Q (SUS)	-1	1	33	-33	-176	176	286	-286
Long. Pm (SUS)	50	50	50	50	50	50	50	50
Long. Pl (SUS)	-3	-3	13	13	-6	-6	14	14
Long. Q (SUS)	-3	3	113	-113	-53	53	85	-85
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	100.0	102.0	100.0	102.0	100.0	102.0	100.0	102.0
Pm+Pl (SUS)	98.0	100.0	110.0	112.0	90.0	92.0	120.0	122.0
Pm+Pl+Q (Total)	97.3	101.3	176.1	129.1	86.1	268.0	406.1	164.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	102.00	138.00	Passed
Pm+Pl (SUS)	122.02	207.00	Passed
Pm+Pl+Q (TOTAL)	406.06	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	5	2	5	2	5	2	5
Circ. Pl (SUS)	-2	-2	10	10	-6	-6	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	0	0	-2	2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	2.0	5.0	2.0	5.0	2.0	5.0	2.0	5.0
Pm+Pl (SUS)	0.0	3.0	12.0	15.0	4.0	1.0	16.0	19.0
Pm+Pl+Q (Total)	0.0	3.0	12.0	15.0	4.0	3.0	16.0	19.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	19.00	207.00	Passed
Pm+Pl+Q (TOTAL)	19.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: SS DRAIN NOZ D1

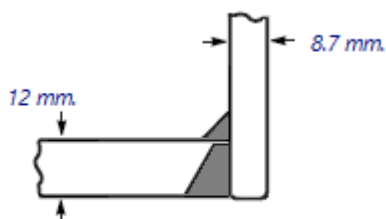
From: 70

Pressure for Reinforcement Calculations	P	25.057	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	850.00	mm.
Design Length of Section	L	2921.6335	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2810.18	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SS DRAIN NOZ D1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.
Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.06 \cdot 426.5 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
= 7.8297 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.06 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
= 0.4209 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3185 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	91.6996 mm.
Parallel to Vessel Wall, opening length	d	45.8498 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: SS DRAIN NOZ D1.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9209 mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.3297 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.3297 mm.
Wall Thickness, shell/head, external pressure	trb2	= 1.8198 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm.
Wall Thickness per table UG-45	tb3	= 4.9200 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[4.92, max(9.3297, 3.0)]
= 4.9200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.9209, 4.92)
= 4.9200 mm.

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: SS DRAIN NOZ D1

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376 mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
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Nozzle Weld $5.0663 = 0.7 * t_{min}$. $7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0717 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 213.0717 mm.

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Input, Nozzle Desc: TEMP INST. T1

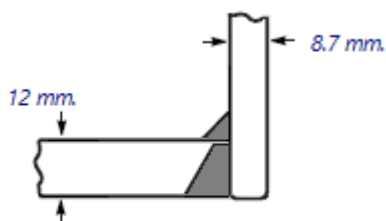
From: 70

Pressure for Reinforcement Calculations	P	25.057	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	850.00	mm.
Design Length of Section	L	2921.6335	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4010.18	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TEMP INST. T1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 1.687 in.
 Actual Thickness Used in Calculation 0.344 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.06 \cdot 426.5 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
 $= 7.8297$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.06 \cdot 22.92 / (138 \cdot 1.0 - 0.6 \cdot 25.06)$
 $= 0.4209$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3185 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	91.6996	mm.
Parallel to Vessel Wall, opening length	d	45.8498	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	18.0940	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: TEMP INST. T1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.9209	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.3297	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.3297	mm.
Wall Thickness, shell/head, external pressure	trb2	= 1.8198	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.9200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.92, \max(9.3297, 3.0)]$
 $= 4.9200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.9209, 4.92)$
 $= 4.9200$ mm.

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46	°C
Calculated Minimum Design Metal Temperature	-104	°C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104	°C

Weld Size Calculations, Description: TEMP INST. T1

Intermediate Calc. for nozzle/shell Welds	Tmin	7.2376	mm.
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Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld $5.0663 = 0.7 * t_{min}$. $7.0700 = 0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0717 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 213.0717 mm.

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Input, Nozzle Desc: LEVEL INST L1

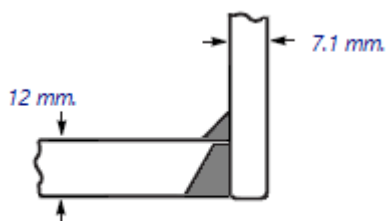
From: 80

Pressure for Reinforcement Calculations	P	25.015	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	850.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	282.8430	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-225.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: LEVEL INST L1



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338 in.
Actual Thickness Used in Calculation	0.281 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $25.02 \cdot 853.0 \cdot 0.897 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.02)$
= 6.9477 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $25.02 \cdot 18.49 / (138 \cdot 1.0 - 0.6 \cdot 25.02)$
= 0.3389 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2796 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	80.9676 mm.
Parallel to Vessel Wall, opening length	d	40.4838 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.0935 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL INST L1.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: LEVEL INST L1

Intermediate Calc. for nozzle/shell Welds	Tmin	5.6374 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.9462 = 0.7 \cdot tmin.$	$7.0700 = 0.7 \cdot Wo \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: LEVEL INST L1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338 in.
Actual Thickness Used in Calculation	0.281 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $25.02 \cdot 853.0 \cdot 0.897 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.02)$



= 6.9477 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $25.02 \cdot 18.49 / (138 \cdot 1.0 - 0.6 \cdot 25.02)$

= 0.3389 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2796 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 73.9704 mm.

Parallel to Vessel Wall, opening length d 36.9852 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 14.0935 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL INST L1.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 1.8389 mm.

Wall Thickness per UG16(b), tr16b = 3.0000 mm.

Wall Thickness, shell/head, internal pressure trb1 = 9.2095 mm.

Wall Thickness tb1 = max(trb1, tr16b) = 9.2095 mm.

Wall Thickness, shell/head, external pressure trb2 = 1.8182 mm.

Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm.

Wall Thickness per table UG-45 tb3 = 4.7200 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[4.72, max(9.2095, 3.0)]

= 4.7200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(1.8389, 4.72)

= 4.7200 mm.

Available Nozzle Neck Thickness = $0.875 \cdot 7.137 = 6.245$ mm. -> OK

Weld Size Calculations, Description: LEVEL INST L1

Intermediate Calc. for nozzle/shell Welds Tmin 5.6374 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.9462 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 9.9367 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.8116 mm.

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Input, Nozzle Desc: LEVEL INST L2

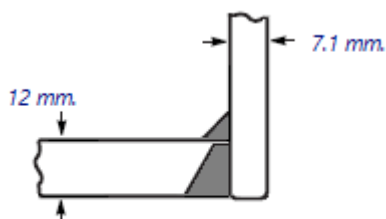
From: 80

Pressure for Reinforcement Calculations	P	25.042	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	850.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	282.8430	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-135.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: LEVEL INST L2



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338 in.
Actual Thickness Used in Calculation	0.281 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 25.04 \cdot 853.0 \cdot 0.897 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.04)$
 $= 6.9551 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.04 \cdot 18.49 / (138 \cdot 1.0 - 0.6 \cdot 25.04)$
 $= 0.3393 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2796 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	80.9676 mm.
Parallel to Vessel Wall, opening length	d	40.4838 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.0935 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL INST L2.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: LEVEL INST L2

Intermediate Calc. for nozzle/shell Welds	Tmin	5.6374 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.9462 = 0.7 \cdot t_{min.}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: LEVEL INST L2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338 in.
Actual Thickness Used in Calculation	0.281 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 25.04 \cdot 853.0 \cdot 0.897 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.04)$

= 6.9551 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $25.04 \cdot 18.49 / (138 \cdot 1.0 - 0.6 \cdot 25.04)$
 = 0.3393 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2796 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	73.9704	mm.
Parallel to Vessel Wall, opening length	d	36.9852	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.0935	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL INST L2.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8393	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.2177	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.2177	mm.
Wall Thickness, shell/head, external pressure	trb2	= 1.8182	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.7200	mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[4.72, max(9.2177, 3.0)]
 = 4.7200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.8393, 4.72)
 = 4.7200 mm.

Available Nozzle Neck Thickness = $0.875 \cdot 7.137 = 6.245$ mm. -> OK

Weld Size Calculations, Description: LEVEL INST L2

Intermediate Calc. for nozzle/shell Welds	Tmin	5.6374	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.9462 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 9.9367 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.8116 mm.

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Input, Nozzle Desc: LEVEL INST L3

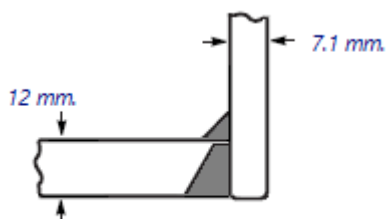
From: 80

Pressure for Reinforcement Calculations	P	25.015	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	850.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	282.8430	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: LEVEL INST L3



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338 in.
Actual Thickness Used in Calculation	0.281 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 25.02 \cdot 853.0 \cdot 0.897 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.02)$
 $= 6.9477$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.02 \cdot 18.49 / (138 \cdot 1.0 - 0.6 \cdot 25.02)$
 $= 0.3389$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2796 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	80.9676 mm.
Parallel to Vessel Wall, opening length	d	40.4838 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.0935 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL INST L3.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: LEVEL INST L3

Intermediate Calc. for nozzle/shell Welds	Tmin	5.6374 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.9462 = 0.7 \cdot tmin.$	$7.0700 = 0.7 \cdot Wo$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: LEVEL INST L3

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338 in.
Actual Thickness Used in Calculation	0.281 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 25.02 \cdot 853.0 \cdot 0.897 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.02)$



= 6.9477 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $25.02 \cdot 18.49 / (138 \cdot 1.0 - 0.6 \cdot 25.02)$
 = 0.3389 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2796 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	73.9704	mm.
Parallel to Vessel Wall, opening length	d	36.9852	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.0935	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL INST L3.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8389	mm.
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.2095	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.2095	mm.
Wall Thickness, shell/head, external pressure	trb2	= 1.8182	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm.
Wall Thickness per table UG-45	tb3	= 4.7200	mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[4.72, max(9.2095, 3.0)]
 = 4.7200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.8389, 4.72)
 = 4.7200 mm.

Available Nozzle Neck Thickness = $0.875 \cdot 7.137 = 6.245$ mm. -> OK

Weld Size Calculations, Description: LEVEL INST L3

Intermediate Calc. for nozzle/shell Welds Tmin 5.6374 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.9462 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 9.9367 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.8116 mm.

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Input, Nozzle Desc: LEVEL INST L4

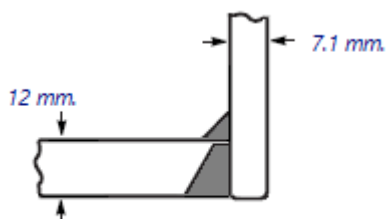
From: 80

Pressure for Reinforcement Calculations	P	25.042	bars
Temperature for Internal Pressure	Temp	85	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	85	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	850.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	12.0000	mm.
Head Internal Corrosion Allowance	c	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	282.8430	mm.
User Entered Minimum Design Metal Temperature		-40.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-45.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: LEVEL INST L4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338 in.
Actual Thickness Used in Calculation	0.281 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 25.04 \cdot 853.0 \cdot 0.897 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.04)$
 $= 6.9551 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 25.04 \cdot 18.49 / (138 \cdot 1.0 - 0.6 \cdot 25.04)$
 $= 0.3393 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.2796 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	80.9676 mm.
Parallel to Vessel Wall, opening length	d	40.4838 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.0935 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL INST L4.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

*Note:
 This Material was specified as being an Impact Tested (Low Temperature) Material.*

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C
Gov. MDMT of the nozzle to shell joint welded assembly :	-104 °C

Weld Size Calculations, Description: LEVEL INST L4

Intermediate Calc. for nozzle/shell Welds	Tmin	5.6374 mm.
---	------	------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.9462 = 0.7 \cdot t_{min.}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: LEVEL INST L4

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.338 in.
Actual Thickness Used in Calculation	0.281 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 25.04 \cdot 853.0 \cdot 0.897 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.04)$



= 6.9551 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $25.04 \cdot 18.49 / (138 \cdot 1.0 - 0.6 \cdot 25.04)$

= 0.3393 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.2796 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 73.9704 mm.

Parallel to Vessel Wall, opening length d 36.9852 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 14.0935 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: LEVEL INST L4.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 1.8393 mm.

Wall Thickness per UG16(b), tr16b = 3.0000 mm.

Wall Thickness, shell/head, internal pressure trb1 = 9.2177 mm.

Wall Thickness tb1 = max(trb1, tr16b) = 9.2177 mm.

Wall Thickness, shell/head, external pressure trb2 = 1.8182 mm.

Wall Thickness tb2 = max(trb2, tr16b) = 3.0000 mm.

Wall Thickness per table UG-45 tb3 = 4.7200 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[4.72, max(9.2177, 3.0)]

= 4.7200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(1.8393, 4.72)

= 4.7200 mm.

Available Nozzle Neck Thickness = $0.875 \cdot 7.137 = 6.245$ mm. -> OK

Weld Size Calculations, Description: LEVEL INST L4

Intermediate Calc. for nozzle/shell Welds Tmin 5.6374 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.9462 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 9.9367 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.8116 mm.

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Input Echo, Tubesheet Number 1, Description: TS

Shell Data:

Main Shell Description: SHELL-1

Shell Maximum Design Pressure	Psd,max	25.00 bars
Shell Maximum Operating Pressure	Psox,max	25.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	12.0000 mm.
Shell Internal Corrosion Allowance	cas	1.5000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	410.000 mm.
Shell Temperature for Internal Pressure	Ts	85.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	25.00 bars
Channel Maximum Operating Pressure	Ptox,max	25.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	12.0000 mm.
Channel Internal Corrosion Allowance	cac	1.5000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	410.000 mm.
Channel Design Temperature	TEMPC	85.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	147
Tube Wall Thickness	et	2.1100 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	3600.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		3545.00 mm.
Design Temperature of the Tubes		85.00 °C
Tube Material		SA-334 6
Tube Material UNS Number		K03006
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		118.00 N./mm ²
Tube Allowable Stress At Ambient		118.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	223.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm.
Groove Weld Leg	ag	2.1100 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	178.962 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	6.0000 mm.

Tubesheet Data:



Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature T 85.00 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	55.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	1.5000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	1.5000 mm.
Tubesheet Outside Diameter	A	538.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	442.00 mm.
Step 1 Depth on the Tubeside	ht1	6.00 mm.
Step 2 Diameter on the Tubeside	dt2	407.00 mm.
Step 2 Depth on the Tubeside	ht2	6.00 mm.
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	442.00 mm.
Step 1 Depth on the Shellside	hs1	6.00 mm.
Step 2 Diameter on the Shellside	ds2	407.00 mm.
Step 2 Depth on the Shellside	hs2	6.00 mm.
Calculated Tubesheet Diameter per Div 2, 4.18.4		442.00 mm.

Note: Tubesheet diameter is now: 442.000 mm. Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	426.151 mm.
Area of the Untubed Lanes	AL	273.2 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	538.000 mm.
Flange Inside Diameter	B	410.000 mm.
Flange Face Outside Diameter	Fod	445.000 mm.
Flange Face Inside Diameter	Fid	410.000 mm.
Gasket Outside Diameter	Go	439.000 mm.
Gasket Inside Diameter	Gi	413.000 mm.
Small end Hub thk.	g0	10.0000 mm.
Large end Hub thk.	g1	13.0000 mm.
Gasket Factor,	m	2.00
Gasket Design Seating Stress	y	17.24 N./mm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	3.1750 mm.
Full face Gasket Flange Option	Program Selects	

Bolting Information:

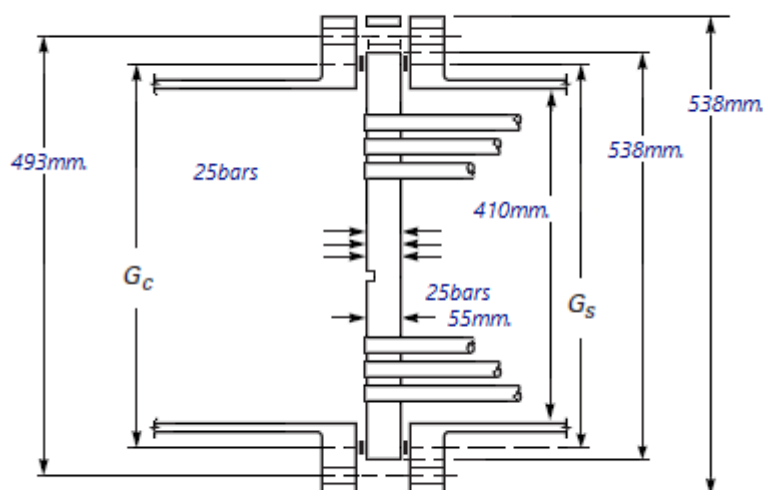
Diameter of Bolt Circle	C	493.000 mm.
Nominal Bolt Diameter	dB	20.0000 mm.
Thread Series	TEMA Metric Thread	
Number of Bolts	n	16
Bolt Material		SA-320 L7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm.

Alternate Flange Operating Bolt Load, Wm1	45206.09 KgF
Alternate Flange Seating Bolt Load, Wm2	15118.27 KgF
Alternate Flange Design Bolt Load, W	60907.91 KgF

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet Tf	31.0000 mm.
Is Bolt Load Transferred to the Tubesheet	Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (25.0 \cdot (19.05/2 - 0)) / (118.0 \cdot 1.0 + 0.4 \cdot 25.0)$
 $= 0.2001 + 0.0000 = 0.2001 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 85.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):



TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	3723.96	9.03	50.0000	0.0134948	122.73

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 192.3322$ bars

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4903	19.05	3723.96	38.85	50.0000	0.0007287	72.86

EMAWP = $(4*B) / (3*(D/T)) = (4 * 72.8551) / (3 * 38.8524) = 25.0009$ bars

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4903 mm.
Allowable Internal Pressure at Corroded thickness	286.79 bars
Required Internal Design Pressure	25.00 bars
Allowable External Pressure at Corroded thickness	192.33 bars
Required External Design Pressure	25.00 bars
Required Thickness due to Shell Side pressure	0.4903 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm.
Basic Gasket Width,	$b0 = N / 2$	6.500 mm.
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	6.422 mm.
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	426.156 mm.

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 * ro + dt$
 $= 2 * 178.962 + 19.05$
 $= 376.974$ mm.

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (25.4 - 19.05) / 25.4$
 $= 0.250$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Gs / Do$ (Configurations d, e, f)
 $= 426.1556 / 376.974$
 $= 1.1305$

Ratio [pc]:

$= Gc / Do$ (Configurations d)
 $= 426.1509 / 376.974$
 $= 1.1305$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$= ltx / h$
 $= 0 / 55.0$
 $= 0.000$ (must be $0 \leq p \leq 1$)

Effective tube hole Diameter [d^*]:

$= \max(dt - 2tt(Et/E) * (StT/S) * (p); dt - 2tt)$
 $= \max(19.05 - 2 * 2.11 (0.19904E+09 / 1.9904E+09) * (117/138) * (0); 19.05 - 2 * 2.11)$



= 19.0500 mm.

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))}$$

$$= 25.4 / \sqrt{(1 - 4 \cdot \min(273.18; 4 \cdot 376.974 \cdot 25.4) / (\pi \cdot 376.974^2))}$$

$$= 29.2274 \text{ mm.}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (29.2274 - 19.05) / 29.2274$$

$$= 0.3482$$

$h/p = 2.165354$; $\mu^* = 0.348215$; $E^* = 68631056 \text{ KPa.}$

$$E^*/E = 0.344806 ; \nu^* = 0.327284$$

Note: As h/p (2.165) is > 2, data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / Do$$

$$= 442.0 / 376.974$$

$$= 1.1725$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (E \cdot \ln(K))$$

$$= (1 - 0.33) / 68631056 \cdot (0.19904E+09 \cdot \ln(1.17))$$

$$= 0.3105$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt]$$

$$= 376.974^2 / 16 \cdot [(1.13 - 1) \cdot (1.13^2 + 1) \cdot 25.0 - (1.13 - 1) \cdot (1.13^2 + 1) \cdot 25.0]$$

$$= 7.1481 \text{ bars} \cdot \text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do)$$

$$= 7.1 + 60907.9 \cdot (426.151 - 426.151) / (2 \cdot \pi \cdot 376.974)$$

$$= 7.1481 \text{ bars} \cdot \text{mm.}^2$$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^* - Do^2 / 32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= (7.15 - 376.974^2 / 32 \cdot 0.31 (25.0 - 25.0)) / (1 + 0.31)$$

$$= 5.4546 \text{ bars} \cdot \text{mm.}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2 / 64 \cdot (3 + \nu^*) \cdot (Ps - Pt)$$

$$= 5.45 + 376.974^2 / 64 \cdot (3 + 0.327) \cdot (25.0 - 25.0)$$

$$= 5.4546 \text{ bars} \cdot \text{mm.}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|)$$

$$= \max(|5.455|; |5.455|)$$

$$= 5.4546 \text{ bars} \cdot \text{mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - hg')^2)$$

$$= 6 \cdot 5.455 / (0.3482 (55.0 - 6.0)^2)$$

$$= 0.0039 \text{ N./mm}^2$$



Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138.0 \\ &= 276.00 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 5.452 / (0.3482 (0.508 - 6.0)^2) \\ &= 0.3115 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 0.508 + 0 + 0 \\ &= 0.5080 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(0.508 ; 0) \\ &= 0.5080 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |25.0 - 25.0| \\ &= 0.000 \text{ bars} \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.25 \cdot 55.0 / 376.97 \cdot \min(0.8 \cdot 138.0 ; 0.533 \cdot 229.6) \\ &= 16.107 \text{ N./mm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.25) \cdot (376.97 / 110.4) \cdot |25.0 - 25.0| + 0 \\ &= 0.0000 \text{ mm.} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.11) 2.11 \\ &= 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.11 (19.05 - 2.11) 117.998 \\ &= 1351.112 \text{ Kg} \end{aligned}$$

Groove Weld Strength [F_g]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 118.0 \\ &= 1351.1116 \text{ Kg} \end{aligned}$$

Weld Strength Factor [f_w]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 117.998 / 117.998 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [F_f]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 118.0 \end{aligned}$$



= 897.6866 Kgf

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right)$$

$$= 2.562 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm.

Minimum Required Groove Weld Leg [agr]: 1.2810 mm.

Tube-Tubesheet Jt allowable, 1351.11 is >= tube strength 1351.11 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	0.0 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	0.0 <=	110.4 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	0.508	55.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	42.125 ...	150	276 17 110	Ps+Pt D1	Ok
D2uc	42.125 25.397	150	276 17 110	Ps+Pt D2	Ok
D3uc	0.508 ...	...	276 ... 110	Ps+Pt D3	Ok
D1c	43.621 ...	159	276 18 110	Ps+Pt-c D1	Ok
D2c	43.622 25.397	159	276 18 110	Ps+Pt-c D2	Ok
D3c	3.508 ...	...	276 ... 110	Ps+Pt-c D3	Ok
Max:	43.6217 25.397 mm.	0.578	0.164 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:



Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	43.622	55.000	Ok
Tubesheet thickness flanged extension :	25.397	31.000	Ok
Tube thickness :	0.490	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	43.278	1.000	43.277	1.000
Tubesheet Shear Stress	152.277	1.000	152.277	1.000
Tube Pressure Stress	286.787	1.000	192.331	1.000
Tubesheet Extension Stress	No Calc	No Calc	33.738	...
Minimum MAWP	43.278		33.738	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	46.055	1.000	46.055	1.000
Tubesheet Shear Stress	161.062	1.000	161.062	1.000
Tube Pressure Stress	286.787	1.000	192.331	1.000
Tubesheet Extension Stress	No Calc	No Calc	33.738	...
Minimum MAPnc	46.055		33.738	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):
 = tubesheet thickness/4
 = 55.0/4
 = 13.750 mm.

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -18 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(25.0/46.05, 25.0/33.74)
 = 0.741

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: